ATTRACTING INVARIANT GRAPHS AND HIGHER DIMENSIONAL PORCUPINE ATTRACTORS

M. ZAJ AND F. H. GHANE

Department Mathematics, Ferdowsi University of Mashhad

Abstract. We study discrete time skew product dynamical systems with higher dimensional fibers and non-uniformly contracting fiber dynamics. For an open class of such systems driven by expanding or solenoidal base maps, we establish a sharp geometric-statistical dichotomy for maximal attractors: either the attractor is a continuous invariant graph, or it is a solenoidal n-porcupine attractor consisting of a measurable invariant graph together with uncountably many embedded C^1 spines.

In the porcupine regime we identify a symbolic branching mechanism inside the blending region that generates topological transitivity and positive topological entropy on the maximal attractor. This provides a rigorous source of chaotic dynamics without assuming uniform hyperbolicity or the existence of Markov partitions.

To describe statistical behaviour under weak and non-uniform contraction, we introduce the notion of fiberwise SRB-like measures. Under natural assumptions, these measures exist uniquely, are ergodic, and attract fiber distributions at an exponential rate. As a consequence, we obtain a robust form of statistical synchronization, whereby typical fiber distributions converge exponentially to a common invariant probability measure despite the geometric complexity of the attractor.

Finally, we establish regularity criteria for invariant graphs, prove C^2 persistence and parameter stability of the geometric and statistical dichotomy, and present a detailed example verifying all hypotheses and illustrating the porcupine geometry, chaotic dynamics, and statistical synchronization.

Contents

1. Introduction 3

1.1. Attractors and Skew Products 7

2010 Mathematics Subject Classification. 37A05; 37C40; 37C70; 37H15; 37D25; 37D30.

Key words and phrases. skew products, maximal attractor, synchronization, invariant graph, entropy.

1.2. Invariant graphs 7
1.3. Partial hyperbolicity 8
1.4. Main results 9
1.5. Main ingredients and organization 10
2. Random iterations of maps 11
2.1. Bounded distortion property for weakly contractive maps 12
2.1.1. Dominated splittings, cone fields, and the strong stable foliation 12
2.1.2. Curves tangent to the central cone field and their iterations 13
2.1.3. Random iterations of central curves 14
2.1.4. Cusp-like regions 15
2.1.5. Random limits of a ball 16
3. Weakly blending regions 19
4. Skew products and geometrical structures of their attracting invariant graphs 22
4.1. Invariant graphs and maximal attractors 22
4.2. Skew products driven by expanding circle maps and their extensions 24
4.3. The subset $\mathcal{A}_{\ell,\mathbb{T}^1} = \mathcal{A}_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$ of $C_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$ 26
4.4. Skew products with high dimensional porcupine graph 27
5. Chaotic Dynamics in the Porcupine Case 33
5.1. A nontrivial spine at uniform scale 33
5.2. Separated contracting images inside the blending region 33
5.3. Nested symbolic branches and separated orbits 34
5.4. Positive topological entropy 35
5.5. The sets $\mathcal{A}_{\ell,\mathbb{T}^1}$ and $\mathcal{A}_{\mathcal{S}_\ell}$ are nonempty 36
6. Statistical Properties and Fiberwise SRB-like Measures 38
6.1. Existence of fiberwise SRB–like measures 39

6.2. Statistical synchronization and uniqueness of invariant measures 40

6.3. Numerical illustration of uniqueness and convergence 43

7. A concrete example verifying hypotheses (i)–(iii) in Corollary 6.4 and Theorem 4.9 43

7.1. Parabolic fiber map 46

7.1.1. Explicit Computations for the Parabolic Map f_P 46

Jacobian of f_P and Eigenvalues of the Linearization 46

Injectivity of f_P on B for Small $a > 0$ 47

Image Condition $f_P(B) \subset B_{r_0}$ for Small $a > 0$ 47

7.2. Verification of contraction / blending 48

7.3. Doeblin property and uniqueness 48

7.4. Lasota–Yorke inequality and spectral gap 48

7.5. Equivariance and conclusion 48

8. An Applied Example and Numerical Verification of Porcupine Attractors 50

8.1. Model Description 50

8.2. Weakly Attracting Pair and Blending Region 51

8.3. Existence of Porcupine Geometry 51

8.4. Chaotic Dynamics on the Attractor 51

8.5. Fiberwise SRB–like Measure and Statistical Synchronization 52

8.6. Numerical Illustration 52

8.7. Discussion 52

9. Discussion 53

10. Conclusion 55

References 55

1. Introduction

Attractors in forced, random, or nonautonomous dynamical systems exhibit a wide range of topological, geometric, and statistical behaviours depending on the perspective adopted, including topological attractors, invariant graph attractors, pullback attractors, and SRB–measure attractors [32, 26]. A unifying and flexible framework for studying such phenomena is provided by discrete time skew products of the form

$$F(\theta, y) = (S(\theta), f_\theta(y)), \qquad F : \Theta \times Y \to \Theta \times Y, \tag{1.1}$$

where (Θ, S) is a compact driving system and each map $f_\theta : Y \to Y$ acts on a compact fiber space Y. Skew products arise naturally in deterministic forcing, random and stochastic dynamics, partially hyperbolic extensions, and time dependent control systems. They capture the interaction between expansion or complexity in the base dynamics and contraction, neutrality, or non-uniform behaviour in the fibers. This interaction is responsible for many of the non-hyperbolic mechanisms observed in modern applied and theoretical dynamics.

Skew products with weak or non-uniform contraction in the fiber direction arise broadly in applications:

- Neuroscience and coupled oscillators. Forced neuron models and slow-fast synaptic systems often combine expanding phase dynamics with multidimensional internal states subject to weak contraction. Complex firing patterns and intermittent synchronization naturally correspond to nontrivial fiber attractors.

- Climate and geophysical models. Quasiperiodically or randomly driven energy–balance and climate models involve multidimensional internal variables with non-uniform contraction. Fiberwise statistical laws describe averaged climate responses, while synchronization reflects alignment of internal climate modes.

- Control systems and network dynamics. Hybrid controllers, symbolic switching protocols, and time dependent routing schemes often exhibit expanding base actions coupled with weakly contractive internal subsystems. Understanding long term behaviour under irregular forcing is essential for robustness and stability.

- Random dynamical systems and filtering. Nonlinear filtering and stochastic forcing frequently induce contraction of internal states, leading to collapse of distributions toward invariant statistical laws. A rigorous description of this convergence requires a fiberwise statistical framework.

These applications motivate the need for a unified theory of skew products with higher dimensional fibers and non-uniform contraction, capable of describing both their geometric structure (graphs, multi graphs, porcupine type attractors) and their statistical behaviour (SRB-like measures and synchronization phenomena).

When the fiber dynamics is uniformly contracting, classical results guarantee the existence of a unique continuous invariant graph attracting all fibers; see Stark [33]. In the case of one-dimensional fibers, finer geometric classifications have been obtained, including pinched attractors, fat solenoids, and bony attractors [21, 34, 25].

From a statistical perspective, the foundational work of Alves–Bonatti–Viana [2] established the existence of SRB measures for partially hyperbolic systems whose central direction is mostly expanding. This theory explains how physical measures persist under strong non-uniformity, but it is intrinsically expansion based and does not address statistical behaviour arising from fiberwise contraction.

Despite these advances, major challenges remain for forced systems and general skew products:

- most existing results either assume $\dim Y = 1$ or rely on strong uniform hyperbolicity;
- non-uniform contraction destroys classical graph regularity and produces geometric structures, such as multi graphs and porcupines, with no one–dimensional analogue;
- higher dimensional fibers lack the order and monotonicity tools available in dimension one;
- existing SRB theories do not describe statistical behaviour in weakly contracting fiber regimes nor explain synchronization in this setting;
- a unified geometric and statistical classification of maximal attractors for nonlinear skew products has been lacking.

Several recent works address parts of this picture, though none provides a complete geometric statistical theory.

(A) Non-uniform geometry and multi–graphs. Rabiee–Ghane–Zaj [31] construct multi graph and bony multi graph attractors, but only for one dimensional fibers and without statistical results or synchronization phenomena.

(B) Non-hyperbolic behaviour in dimension one. Barrientos–Cisneros [5] and Barrientos–Chávez [4] demonstrate robust zero Lyapunov exponents and historical behaviour in transitive skew products, again restricted to one dimensional fibers.

(C) Nonautonomous and fractional order systems. Cui–Kloeden [15] establish existence of bounded skew product attractors for fractional equations, but do not address nonlinear geometric classification or SRB type statistics.

(D) Invariant distributions and fractal graphs. Xu–Zhang [37] study invariant distributions and fractal graph structures, without considering attractor classification or fiberwise statistics.

(E) Empirical and variational approaches. Díaz et al. [17] and Crovisier et al. [14] provide deep insights into ergodic measures and empirical distributions under weak hyperbolicity, but not specifically for skew products with heterogeneous forcing.

(F) SIAM-line advances in synchronization and linear theory. Recent SIAM contributions [22, 7, 6] develop synchronization criteria and angular-value theory for nonautonomous linear systems. These tools do not address nonlinear geometric dichotomies or statistical synchronization in skew products.

This paper develops a unified geometric and statistical theory for skew products with

- fibers of dimension $\dim Y \geq 2$,
- non-uniform weak contraction in the fiber direction,
- expanding or solenoidal base dynamics.

Our main contributions are as follows:

(i) Geometric dichotomy for maximal attractors. We prove a sharp dichotomy: for an open class of skew products, the maximal attractor is either (a) a continuous invariant graph, or (b) a solenoidal n–porcupine attractor consisting of a measurable invariant graph together with uncountably many embedded C^1 spines. This provides the first such classification for higher dimensional, non-uniformly contracting fibers.

(ii) Chaotic dynamics inside porcupine attractors. In the porcupine regime we prove the existence of symbolic branching mechanisms that generate topological transitivity and positive topological entropy on the maximal attractor. This establishes the presence of chaotic dynamics without assuming uniform hyperbolicity or Markov partitions.

(iii) Fiberwise SRB-like measures. We introduce the notion of fiberwise SRB-like measures describing statistical behaviour inside weakly contracting fibers. We prove existence, uniqueness, ergodicity, and exponential convergence, extending the SRB program of [2] to contraction dominated skew products.

(iv) Statistical synchronization. Despite the geometric complexity of porcupine attractors, fiber distributions converge exponentially to the unique fiberwise SRB-like measure. This phenomenon, which we term statistical synchronization, has no analogue in one dimensional geometric studies and no counterpart in expansion based SRB theory.

(v) Regularity, robustness, and examples. We establish regularity criteria for invariant graphs, analyze the breakdown of regularity under loss of uniform contraction, prove C^2 persistence of the dichotomy, and provide a detailed example verifying all hypotheses.

1.1. Attractors and Skew Products. We start recalling some classical definitions related to attractors. Many variants for the definition of an attractor can be found in the literature, see [28] for a discussion. In what follows, let M be a compact, connected, closed, and smooth Riemannian manifold, m the normalized Lebesgue measure on M, and $\psi\colon M \to M$ a continuous map.

Given a closed ψ-invariant set A, its basin of attraction $B(A)$ is the set of points whose future orbits tend (in the Hausdorff metric) to A. We say that A is Milnor attractor for ψ if $m(B(A)) > 0$ [28]. In some definitions of attractors, the basin of attraction $B(A)$ is an open set. However, here we only require a basin of attraction with positive measure.

A compact set D is called trapping for ψ if $\psi(D) \subset \operatorname{int}(D)$, where $\operatorname{int}(D)$ denotes the interior of D. In such a case, the closed ψ-invariant set

$$A_{max}(D, \psi) := \bigcap_{n\geq 0} \psi^n(D) \tag{1.2}$$

is the maximal attractor for ψ in D.

In discrete time, random dynamical systems influenced by a deterministic external factor are modeled as skew product systems. Consider compact metric spaces Θ and Y and a continuous map $S\colon \Theta \to \Theta$. Fixed S, we denote by $C_{S,\mathrm{Lip}}(\Theta \times Y)$ the set of skew products

$$F = F_{(S,f_\theta)}\colon \Theta \times Y \to \Theta \times Y, \qquad F(\theta, y) = (S(\theta), f_\theta(y)), \tag{1.3}$$

whose fiber maps f_θ, $\theta \in \Theta$, are Lipschitz and continuously depend on θ. Here, S denotes the base transformation and f_θ represents the fiber transformation (or forced system).

The fiber map f_θ can be considered as a map from Y to Y, where Y is the fiber space. We adopt the usual notation $F^n(\theta, y) = (S^n\theta, f_\theta^n(y))$, for the iterates F^n of F, where $f_\theta^n = f_{S^{n-1}(\theta)} \circ \cdots \circ f_\theta$. Hence, $f_\theta^{n+k}(y) = f_{S^k\theta}^n(f_\theta^k(y))$. If the fiber maps are invertible, for $n = 1$ and $k = -1$ this includes the identity and $f_\theta^{-1}(y) = (f_{S^{-1}\theta})^{-1}(y)$.

1.2. Invariant graphs. We now discuss invariant graphs and their variations. In the study of skew products, invariant graphs, particularly attracting invariant graphs, play an essential role. In fact, they are the natural substitutes of fixed points and considerably simplify the dynamics of the forced systems.

Definition 1.1. Let m be a S-invariant measure. A measurable function $\gamma : \Theta \to Y$ is

i) an F-invariant graph if for every $\theta \in \Theta$ it holds

$$F(\theta, \gamma(\theta)) = (S(\theta), \gamma(S(\theta))), \quad \text{or} \quad f_\theta(\gamma(\theta)) = \gamma(S(\theta)),$$

ii) an (F, m)-invariant graph if

$$f_\theta(\gamma(\theta)) = \gamma(S(\theta)), \quad \text{for } m-a.e.\ \theta \in \Theta. \tag{1.4}$$

With slight abuse of notation, we call the point set

$$\Gamma = \Gamma_\gamma := \{(\theta, \gamma(\theta)) : \theta \in \Theta\}$$

F or (F, m)-invariant graph as well, but labeled with the corresponding capital letter.

The graph Γ is an attracting invariant graph if for m-a.e. $\theta \in \Theta$, one has that

$$\lim_{n\to\infty} |\pi_y(F^n(\theta, y)) - \gamma(S^n(\theta))| = 0,$$

for all $y \in Y$.

In skew products, Lyapunov exponents are useful quantitative indicators of the asymptotic expansion and contraction rates (along fibers).

Definition 1.2 (Maximal Lyapunov exponent). Let $F : \Theta \times Y \to \Theta \times Y$ be a skew product whose fiber maps f_θ are differentiable. The maximal Lyapunov exponent of an F-invariant probability measure μ is defined by

$$\lambda(\mu, F) := \lim_{n\to\infty} \frac{1}{n} \int_{\Theta\times Y} \log \|Df_\theta^n(x)\| \, d\mu(\theta, x). \tag{1.5}$$

The maximal Lyapunov exponent of an (F, m)-invariant graph γ is defined by

$$\lambda_m(\gamma) := \int_\Theta \log \|Df_\theta(\gamma(\theta))\| \, dm(\theta). \tag{1.6}$$

1.3. Partial hyperbolicity. Consider a smooth manifold M (maybe noncompact), a differentiable dynamical system $f : M \to M$ and an f-invariant set Λ. A splitting $T_\Lambda M = E \oplus F$ is dominated if E and F are Df-invariant and there are constants $m > 0$ and $K < 1$ such that

$$\|(D_x f^m)|_{E_x}\| \|((D_x f^m)^{-1})|_{F_{f^m(x)}}\| < K, \text{ for all } x \in \Lambda. \tag{1.7}$$

A Df-invariant bundle E defined on Λ is uniformly contracting (resp. expanding) if there are $C > 0$ and $0 < \lambda < 1$ such that for every $n > 0$ one has

$$\|D_x f^n(v)\| \le C\lambda^n \|v\| \ \ (\text{resp. } \|D_x f^{-n}(v)\| \le C\lambda^n \|v\|) \text{ for all } x \in \Lambda \text{ and } v \in E. \tag{1.8}$$

We say that the set Λ is partially hyperbolic if there is a dominated splitting $E \oplus F$ of $T_\Lambda M$ such that either E is uniformly contracting or F is uniformly expanding. In the first case we write $T_\Lambda M = E^{ss} \oplus E^{cu}$, otherwise we write $T_\Lambda M = E^{uu} \oplus E^{cs}$.

An f-invariant set Λ is strongly partially hyperbolic if there is a df-invariant dominated splitting $E^s \oplus E^c \oplus E^u$ where $df_{|E^s}$ is uniformly contracting, $df_{|E^u}$ is uniformly expanding, E^c is non-trivial and non-hyperbolic, and the three bundles E^s, E^c, and E^u are non-trivial.

1.4. Main results. The space of all skew products $F_{(\ell, f_t)}$ as in (4.9) is denoted by $\mathcal{C}_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$. We denote by $\mathcal{C}(\mathcal{S}_\ell \times \mathbb{B}_n)$ the space of skew product maps $\mathbb{F} = \mathbb{F}_F$ which are an extension of some $F \in \mathcal{C}_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$. We equip these spaces with the metric given by (4.12)

For the skew product with trapping region D, the set $\Lambda := A_{max}(D)$ is the maximal attractor. Given any point a in the base space let $\Lambda_a := \Lambda \cap \{a\} \times \mathbb{B}_n$. We call Λ_a a spine and say that it is non-trivial if it is not a singleton. Note that for any a in the base space, $\Lambda = \bigcup \Lambda_a$.

The following concept is a generalization of porcupine horseshoes discovered by Diaz and Gelfert in [16]. Now, we define Solenoidal porcupine-like graph as follows:

Definition 1.3 (Solenoidal porcupine-like graph). Given $\mathbb{F} \in C(\mathcal{S}_\ell \times \mathbb{B}_n)$, a compact $\mathbb{F}$-invariant set Λ is a solenoidal n-porcupine-like graph if

(1) Λ is the maximal attractor for $\mathbb{F}$ in some trapping region D;

(2) there exist a subset $\widetilde{\mathcal{S}}_\ell$ of $\mathcal{S}_\ell$ with $\nu_\ell(\widetilde{\mathcal{S}}_\ell) = 1$ and an $(\mathbb{F}, \nu_\ell)$-invariant (attracting) graph $\gamma_{\mathbb{F}}$ defined on $\widetilde{\mathcal{S}}_\ell$ such that Λ is the union of the graph $\Gamma_{\gamma_{\mathbb{F}}}$ and a set consisting of uncountably many one dimensional C^1 closed submanifold contained in the closure of $\Gamma_{\gamma_{\mathbb{F}}}$ (called nontrivial spines). The map $\gamma_{\mathbb{F}}$ is called the graph function of the maximal attractor.

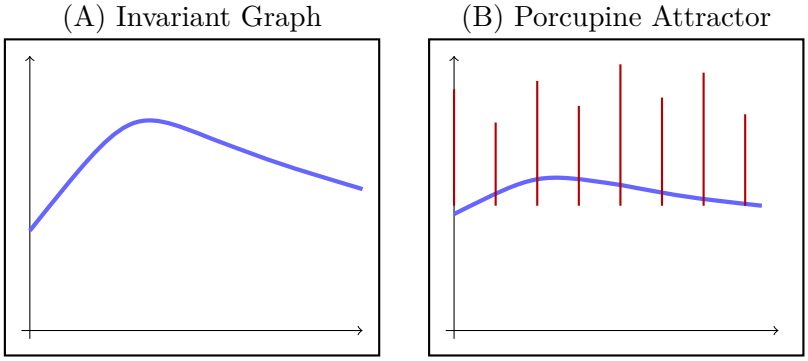


Figure 1. Schematic representation of the geometric dichotomy for maximal attractors in skew products with non-uniformly contracting fibers. In the absence of a continuous invariant graph, the maximal attractor exhibits a solenoidal porcupine structure, consisting of a measurable invariant graph (the central core) decorated by uncountably many embedded C^1 spines. Each spine arises from stable remnants of random fiber compositions and varies measurably with the base point. The figure illustrates how expansion in the base, combined with weak contraction and parabolic behavior in the fibers, destroys global graph regularity while preserving a compact attracting set, as described in Theorem 4.9.

Here we have three main results as follows:

- Let f be a C^2 weakly contractive injective map and $\mathcal{N}_f$ be a normally hyperbolic adapted C^2 neighborhood of f as in Proposition 2.6. Then for every sequence $\mathbf{t} = (\cdots, t_{-n}, \cdots, t_{-1}, t_0)$ with values in I, such that for each n, $f_{t_{-n}} \in \mathcal{N}_f$ or $f_{t_{-n}}$ is a C^2 injective map with $f_{t_{-n}}(\mathbb{B}_n) \subset \operatorname{int}(\mathbb{B}_n)$, the intersection

$$\Gamma_{\mathbf{t}} = \bigcap_{k\geq 0} \mathbb{B}_{\mathbf{t}}(k), \qquad \mathbb{B}_{\mathbf{t}}(k) := f_{\mathbf{t}}^k(\mathbb{B}_n)$$

is either a single point or a one dimensional C^1 closed submanifold.

- There is an open neighbourhood $\mathcal{V}_{\mathcal{S}_\ell}$ in $\mathcal{C}(\mathcal{S}_\ell \times \mathbb{B}_n)$ such that every $\mathbb{F} \in \mathcal{V}_{\mathcal{S}_\ell}$ has a maximal attractor $A_{max}(\mathbb{F})$ that is topologically transitive and satisfies the following dichotomy:
  - either $A_{max}(\mathbb{F})$ is an attracting continuous invariant graph;
  - or $A_{max}(\mathbb{F})$ is a solenoidal n-porcupine-like graph.
- For a weakly attracting pair $(f_1, \dots, f_k)$ (as defined in definition 3.2) with blending region B, and that the base map $T\colon M \to M$ preserves an ergodic absolutely continuous probability ν, the skew product $F(a, y) = (T(a), f_a(y))$ admits a fiberwise SRB-like measure. In addition this fiberwise SRB-like measure is unique and ergodic.

1.5. Main ingredients and organization. We will provide an overview of the paper's contents and briefly outline the main ideas used to prove the theorem. In Section 3, we introduce two local tools, namely the weakly attracting pair and the weakly blending region of $\mathbb{R}^n$, which allow us to construct n-porcupine attractors. To construct the weakly blending regions of $\mathbb{R}^n$, we consider contractive maps with non-uniform contraction rates, also known as weakly contractive maps. For a specific class of weakly contractive maps, the approach presented in [2] yields a bounded distortion property for random compositions of fiber maps. Partial hyperbolicity is a crucial element in our approach. Furthermore, we demonstrate that under small perturbations of a weakly contractive map, the forward random iterations of the ball $\mathbb{B}_n \subset \mathbb{R}^n$ are either a single point or a C^1-submanifold (as shown in Proposition 2.6). Based on these findings, we investigate the existence of a weakly blending region for each compact ball $\mathbb{B}_n \subset \mathbb{R}^n$, as detailed in Theorem 3.3. Section ?? is dedicated to studying skew products that have non-uniform contraction rates along the fibers. This particular condition enables us to utilize [33, Thm. 1.4] to establish attracting invariant graphs. In Section 4, we complete the proof of Theorem 1 and introduce an open set $\mathcal{V}$ that meets the requirements outlined in Theorem 1. The maps within $\mathcal{V}$ have average fiber contraction rates, and their fiber maps exhibit a weakly blending region. These are the primary ingredients involved in constructing n-porcupine attractors.

2. Random iterations of maps

In this section, we consider contractive maps with non-uniform contraction rates. Following [1], given a metric space (X, d) we say that a continuous map $f\colon X \to X$ is weakly contractive if $d(f(x), f(y)) < d(x, y)$ for every $x, y \in X$ with $x \neq y$. It is well-known, see [21,

Coro. 3] and [8], that if f is weakly contractive and (X, d) is compact then f has a unique fixed point $x \in X$ which is attracting in the sense that $\lim_{k\to\infty} f^k(y) = x$, for every $y \in X$. Here, we take $X = \mathbb{B}_n$, where $\mathbb{B}_n$ denotes the closed unit ball of $\mathbb{R}^n$. Thus, for a weakly contractive map $f\colon \mathbb{B}_n \to \mathbb{B}_n$, we can speak of its attracting fixed point, denoted by x_f.

A key ingredient in our construction of weak blending regions is the following:

Definition 2.1 (Parabolic attracting fixed point). Let $f\colon \mathbb{B}_n \to \mathbb{B}_n$ be a smooth weakly contractive map. The attracting fixed point x_f of f is parabolic if the Jacobian matrix $Df(x_f)$ possesses 1 as a simple eigenvalue and the other eigenvalues have absolute values less than 1. The parabolic fixed point x_f is generic whenever the eigenvalues of $Df(x_f)$ are all positive and of multiplicity one.

In what follows, we denote by $\mathbb{WC}^2(\mathbb{B}_n)$ the set of C^2 weakly contractive injective maps $f\colon \mathbb{B}_n \to \mathbb{B}_n$ with $f(\mathbb{B}_n) \subset \mathrm{int}(\mathbb{B}_n)$ and whose attracting fixed point x_f is generic and parabolic. Note that x_f belongs to the interior $\mathrm{int}(\mathbb{B}_n)$ of $\mathbb{B}_n$. There is a basis of neighborhoods $V_{n,f}$ of x_f with $f(\overline{V_{n,f}}) \subset V_{n,f}$. We call these neighborhoods attracting.

2.1. Bounded distortion property for weakly contractive maps. Following the approach in [2, 38], for maps $f \in \mathbb{WC}^2(\mathbb{B}_n)$, we will provide a bounded distortion property on a region around its attracting fixed point x_f. We start with some preparatory results.

2.1.1. Dominated splittings, cone fields, and the strong stable foliation. Given $f \in \mathbb{WC}^2(\mathbb{B}_n)$, consider the Df-invariant splitting $E^{ss} \oplus E^c$, corresponding to the eigenspaces of $Df(x_f)$ associated to the eigenvalues less than 1 and equal to 1, respectively. Note that E^c is one-dimensional. As this splitting is dominated, we can extend it to a continuous splitting (also denoted by $E^{ss} \oplus E^c$) defined on a small attracting neighborhood V_f of x_f. Note that this new splitting is not necessarily invariant.

For notational convenience, we will use the adapted metric (equivalent to the initial one) provided by [18, Theorem 1]. This allows us to take the constant C in equation (1.8) equal to 1. Hence, there exists $0 < \lambda < 1$ such that

i) $\|Df|_{E^{ss}_x}\| \le \lambda$, for all $x \in \overline{V_f}$;

ii) $\|Df|_{E^{ss}_x}\| \cdot \|Df^{-1}|_{E^c_{f(x)}}\| \le \lambda < 1$ for all $x \in \overline{V_f}$.

Recalling that x_f is parabolic, by shrinking V_f and slightly increasing $\lambda < 1$ if necessary, we have the following strong form of domination

$$\|Df|_{E^{ss}_x}\| \le \lambda \quad \text{and} \quad \|Df|_{E^{ss}_x}\| \cdot \|Df^{-1}|_{E^c_{f(x)}}\|^{1+1} \le \lambda < 1, \tag{2.1}$$

for all $x \in \overline{V_f}$.

For each $0 < \beta < 1$, we define the center cone field $C^c_\beta := (C^c_\beta(y))_{y\in\overline{V_f}}$ of width β by

$$C^c_\beta(y) := \{v_1 + v_2 \in E^{ss}_y \oplus E^c_y : \|v_1\| \le \beta\|v_2\|\}.$$

The strong stable cone field $C^{ss}_\beta := (C^{ss}_\beta(y))_{y\in V_f}$ of width β is analogously defined.

The center cone field is positively invariant, that is $Df(y)(C^c_\beta(y)) \subset C^c_\beta(f(y))$, provided that V_f is small and $y, f(y) \in \overline{V_f}$. Indeed, according to (ii) above,

$$Df(x_f)(C^c_\beta(x_f)) \subset C^c_{\lambda\beta}(x_f) \subset C^c_\beta(x_f).$$

By continuity, this extends (for small V_f) to each $y \in \overline{V_f} \cap f^{-1}(\overline{V_f})$.

By [10, 19], the domination property and the uniform contraction of E^{ss} imply that there is a (uniquely defined) continuous family $\mathcal{F}^{ss}$ of injectively C^2 immersed submanifolds $\{\mathcal{F}^{ss}(x)\colon x \in V_f\}$ of dimension $(n-1)$ such that $x \in \mathcal{F}^{ss}(x)$ and $\mathcal{F}^{ss}(x)$ is tangent to E^{ss}_x at x. This family is invariant, in the sense that $f(\mathcal{F}^{ss}(x)) \subset \mathcal{F}^{ss}(f(x))$ for every $x \in V_f \cap f^{-1}(V_f)$, and the leaves $\mathcal{F}^{ss}(x)$ are uniformly contracted by f. The lamination $\mathcal{F}^{ss}$ is called strong stable foliation and the sets $\mathcal{F}^{ss}(x)$ are called strong stable leaves.

Since the central bundle E^c is one-dimensional, there always exist curves tangent to E^c passing through every point in V_f. These curves are called central leaves and denoted by $\mathcal{F}^c(x)$. In general, these central curves are neither invariant nor uniquely defined. We now focus on these central curves and their iterations.

2.1.2. Curves tangent to the central cone field and their iterations. The next remark is a standard consequence of the persistence of domination and invariant bundles, see for instance [9, Section B1].

Remark 2.2. Consider $f \in \mathbb{WC}^2(\mathbb{B}_n)$, its attracting parabolic fixed point x_f, and a small attracting neighbourhood V_f of x_f. After increasing $\lambda < 1$ and shrinking V_f, we can assume that the dominated splitting $E^{ss} \oplus E^c$ is defined on $\overline{V_f}$ and satisfies (2.1) and there is small $\beta > 0$ such that

$$\|Df(y)v^{ss}\| \cdot \|Df^{-1}(f(y))v^c\| \le \lambda\|v^{ss}\| \cdot \|v^c\|, \tag{2.2}$$

for every pair of vectors $v^{ss} \in C^{ss}_\beta(y)$, $v^c \in C^c_\beta(y)$ and every point $y \in \overline{V_f} \cap f^{-1}(\overline{V_f})$. Moreover, we can assume that there exists a C^2-neighborhood $\mathcal{N}_f$ of f consisting of maps g such that $g(\overline{V_f}) \subset V_f$, the domination conditions in (2.1) hold, $\|Dg|_{E^{ss}_x}\| \le \lambda$, the central cone field is Dg-invariant, i.e., $Dg(y)C^c_\beta(y) \subset C^c_\beta(g(y))$, and

$$\|Dg(y)v^{ss}\| \cdot \|Dg^{-1}(g(y))v^c\| \le \lambda\|v^{ss}\| \cdot \|v^c\|, \tag{2.3}$$

for every $v^{ss} \in C^{ss}_\beta(y)$, $v^c \in C^c_\beta(y)$, and every $y \in \overline{V_f} \cap g^{-1}(\overline{V_f})$.

As a consequence, the maximal invariant set of g in $\overline{V_f}$ is partially hyperbolic with one dimensional central bundle and $(n-1)$-strong stable direction.

In what follows, we fix $f \in \mathbb{WC}^2(\mathbb{B}_n)$, neighbourhoods V_f of x_f and $\mathcal{N}_f$ of f, and cone fields C^{ss}_β and C^c_β satisfying Remark 2.2.

A closed C^1-curve $N \subset V_f$ is tangent to the central cone field C^c_β if for every $x \in N$ the tangent space T_xN is contained in C^c_β. We call these curves central ones. A special case of central curves are the central leaves. Given a central curve N, the invariance conditions in Remark 2.2 imply that for every $g \in \mathcal{N}_f$ we have that $g(N) \subset V_f$ and $g(N)$ is tangent to C^c_β. Inductively, we have that if $g_1, \dots, g_n \in \mathcal{N}_f$ then $g_n \circ \cdots \circ g_1(N) \subset V_f$ and the curve $g_n \circ \cdots \circ g_1(N)$ is a central curve.

Similarly, we define $(n-1)$ disks tangent to the strong stable cone field and call them strong stable disks. Applying Remark 2.2, we have that if $g \in \mathcal{N}_f$ and D is a strong stable disk such that $D \subset V_f$ and $g^{-1}(D) \subset V_f$ then $g^{-1}(D)$ is a strong stable disk with $\operatorname{diam}(D) \le \lambda \operatorname{diam}(g^{-1}(D))$. Inductively, if $g_1, \dots, g_n \in \mathcal{N}_f$ and $D \subset V_f$ is a strong stable disk such that $g_i^{-1} \circ \cdots \circ g_1^{-1}(D) \subset V_f$ for every $i = 1, \dots, n$, then $g_i^{-n} \circ \cdots \circ g_1^{-1}(D)$ is a strong stable disk such that

$$\operatorname{diam}(D) \le \lambda^n \operatorname{diam}(g_i^{-n} \circ \cdots \circ g_1^{-1}(D)). \tag{2.4}$$

Finally, given a set A and a point $x \in A$, we denote by $C(x, A)$ the connected component of A containing x. We will consider the family of strong stable disks $\mathcal{F}^{ss}_{\mathrm{loc}}(x) := C(x, \mathcal{F}^{ss}(x) \cap V_f)$.

2.1.3. Random iterations of central curves. In this subsection, we closely follow the constructions in [2, Section 2.1] and omit some details. Given a central curve N as above, $x \in N$, and $\epsilon > 0$, let $U_x(\epsilon)$ be the open ϵ-neighborhood of x in $\mathbb{B}_n$ and $N_x(\epsilon) := U_x(\epsilon) \cap N$. There is small $r > 0$ such that for each $x \in N$ there is a uniquely defined family of linear maps

$$\mathcal{A}_x(y) \colon T_xN_x \to E^{ss}_x, \quad y \in N_x(r),$$

such that the tangent spaces T_yN_x are parallel to the graphs of $\mathcal{A}_x(y)$.

For given constants $C > 0$ and $0 < \zeta \le 1$, we say that the tangent bundle of N is (C, ζ)-Hölder if there is $r > 0$ such that

$$\|\mathcal{A}_x(y)\| \le C\rho(x, y)^\zeta \quad \text{for every } x \in N \text{ and } y \in N_x(r), \tag{2.5}$$

where $\rho(x,y)$ is the length of the curve connecting x to y in $N_x(r)$. For $0<\zeta\le 1$ we let

$$\kappa_\zeta(N) := \inf\{C>0 : \text{the tangent bundle of } N \text{ is } (C,\zeta)\text{-Hölder}\}. \tag{2.6}$$

In [2] a Hölder exponent ζ is selected satisfying

$$\|Df|_{E^{ss}_x}\| \cdot (\|Df^{-1}|_{E^c_{f(x)}}\|)^{1+\zeta} \le \lambda < 1.$$

By equation (2.1), we can take $\zeta = 1$ and write $\kappa_1(N) = \kappa(N)$. Next proposition is just a reformulation of [2, Corollary 2.4] in our context of random compositions. For that let us introduce some notation: given a sequence $\mathbf{t} = (\cdots, t_{-j}, \cdots, t_{-1}, t_0)$ and maps $f_{t_{-i}}\colon \mathbb{B}_n \to \mathbb{B}_n$, for each $j \ge 0$ we write

$$f^{j+1}_{\mathbf{t}} := f_{t_0} \circ f_{t_{-1}} \circ \cdots \circ f_{t_{-j}}. \tag{2.7}$$

Proposition 2.3 (Corollary 2.4 in [2]). Let $f \in \mathbb{WC}^2(\mathbb{B}_n)$. There are small neighborhoods V_f of x_f and $\mathcal{N}_f$ of f, a constant $\beta > 0$ satisfying Remark 2.2, and a constant $C_1 > 0$ such that for every C^1-curve $N \subset V_f$ tangent to the centre cone field $\mathcal{C}^c_\beta$, every continuous path $t \in I \mapsto f_t \in \mathcal{N}_f$, with $I \subset \mathbb{R}$ an interval, and every sequence $\mathbf{t} = (\cdots, t_{-j}, \cdots, t_{-1}, t_0)$ with values in I the following holds:

a) there exists $n_0 \ge 0$ such that $\kappa(f^n_{\mathbf{t}}(N)) \le C_1$ for every $n \ge n_0$,

b) if $\kappa(N) \le C_1$ then the same is true for every $\kappa(f^n_{\mathbf{t}}(N))$, $n \ge 0$,

c) in particular, if N is as in b), then the Jacobian functions

$$J^n_{\mathbf{t},N}\colon f^n_{\mathbf{t}}(N) \ni y \mapsto \log|\det(Df|_{T_y f^n_{\mathbf{t}}(N)})|, \quad n \ge 0,$$

are $(L,1)$-Hölder continuous, for some $L > 0$ depending only on C_1 and f.

A special case of Proposition 2.3 is the non-random case where $f_{t_{-i}} = g \in \mathcal{N}_f$ for every i.

2.1.4. Cusp-like regions. Consider $f \in \mathbb{WC}^2(\mathbb{B}_n)$, a center leaf $\mathcal{F}^c(x_f)$ passing throughout x_f, and a closed fundamental domain D^c of $\mathcal{F}^c(x_f)$ with boundary points $a \in V_f$ and $f(a)$. For each $x \in D^c$ consider a small closed strong stable disk $D^{ss}(x) \subset \mathcal{F}^{ss}(x)$ containing the point x in its interior and contained in V_f. We choose these disks depending continuously on x and such that $D^{ss}(f(a)) := f(D^{ss}(a))$.

Denote by $C^2(D^c, V_f)$ the space of C^2 maps from D^c to V_f. Let $H\colon D^{ss}(a) \to C^2(D^c, V_f)$ be a continuous map which associates to each $x \in D^{ss}(a)$ a C^2 arc $I_x := H(x)(D^c)$ connecting the points x and $f(x)$ such that $H(a)$ is the identity and hence $I_a = D^c$. Thus, by shrinking

the disks $D^{ss}(x)$, if necessary, we can assume that the curve I_x is a central curve for every $x \in D^{ss}(a)$. Consider now the closed subset

$$W_1 := \bigcup_{x \in D^{ss}(a)} I_x \subset V_f \tag{2.8}$$

foliated by central curves and whose boundary contains the disks $D^{ss}(a)$ and $D^{ss}(f(a))$, see Figure 2.

By construction, the subsets W_1 and $f(W_1)$ are C^2 submanifolds contained in V_f. Arguing recursively, we define the cusp-like region W as follows:

$$W := \bigcup_{k=1}^{\infty} W_k, \quad \text{where } W_k := f(W_{k-1}), \quad k \geq 2. \tag{2.9}$$

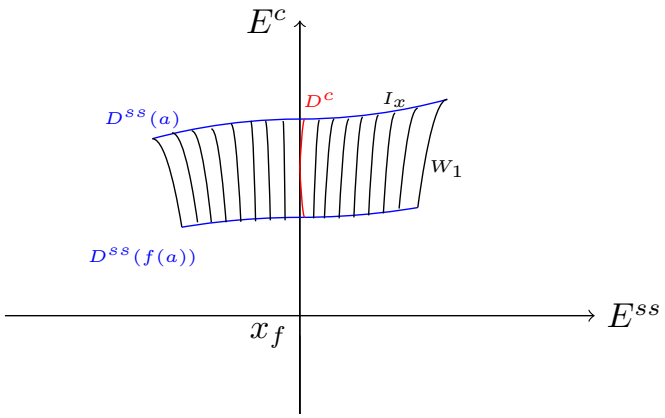


Figure 2. The closed rectangle W_1.

Remark 2.4. Note that $\mathsf{diam}(W_k)$ tends to zero as $k \to \infty$ and that the parabolic point x_f satisfies $x_f \in \overline{W} \setminus W$. Finally, the subsets W_k and W are C^2 submanifolds.

Taking $f_{t_{-i}} = f$ for all i in Proposition 2.3, we have that the tangent bundle of the iterates $f^n(I_x)$ are Hölder continuous with uniform Hölder constants depending only on f and $\kappa_1(D^c)$, see (2.6). Moreover, by continuity of H and compactness of I_x and $D^{ss}(a)$, we have that $\kappa_1(I_x)$ is uniformly bounded independently of x. In particular, by item (c) of Proposition 2.3,

$$J^k_{I_x} \colon f^k(I_x) \ni y \mapsto \log |\det(Df|_{T_y f^k(I_x)})| \tag{2.10}$$

is $(L, 1)$-Hölder continuous, for some constant $L > 0$ that does not depend on k and x.

The observations above are summarized in the next proposition, which is a counterpart of [38, Thm. 2.7] in the n-dimensional case.

Proposition 2.5. Let $f \in \mathbb{WC}^2(\mathbb{B}_n)$ and consider sufficiently small cusp-like region W defined as in (2.9). Then the function

$$J : W \ni y \mapsto \log|\det(Df|_{T_y(W)})|$$

is $(L, 1)$-Hölder continuous, for some constant $L > 0$.

2.1.5. Random limits of a ball. In this section, we prove the following result about forward random iterations of the ball $\mathbb{B}_n$.

Proposition 2.6. Let $f \in \mathbb{WC}^2(\mathbb{B}_n)$. There is a C^2 neighborhood $\mathcal{N}_f$ of f such that for every continuous path $t \in I \mapsto f_t \in \mathcal{N}_f$ and every sequence $\mathbf{t} = (\cdots, t_{-n}, \cdots, t_{-1}, t_0)$ with values in I, the intersection

$$\Gamma_{\mathbf{t}} = \bigcap_{k\geq 0} \mathbb{B}_{\mathbf{t}}(k), \qquad \mathbb{B}_{\mathbf{t}}(k) := f^k_{\mathbf{t}}(\mathbb{B}_n)$$

is either a single point or a one dimensional C^1 closed submanifold.

Proof. The definition of $f^k_{\mathbf{t}}$ in (2.7) implies that $(\mathbb{B}_{\mathbf{t}}(k))_k$ is a nested sequence of compact and connected sets, therefore its intersection $\Gamma_{\mathbf{t}}$ is compact and connected.

Take neighborhoods V_f of x_f and $\mathcal{N}_f$ of f as in Remark 2.2. Fix a center leaf $\mathcal{F}^c(x_f)$ passing throughout x_f and let $\mathcal{F}^c_{\text{loc}}(x_f) = C(x_f, \mathcal{F}^c(x_f) \cap V_f)$. We consider the neighborhood of x_f given by

$$V = \bigcup_{x\in\mathcal{F}^c_{\text{loc}}(x_f)} \mathcal{F}^{ss}_{\text{loc}}(x) \subset V_f, \tag{2.11}$$

where $\mathcal{F}^{ss}_{\text{loc}}(x) := \mathcal{F}^{ss}(x) \cap V_f$. By shrinking V_f and $\mathcal{N}_f$, we can assume that $g(\overline{V}) \subset \text{int}(V)$ for all $g \in \mathcal{N}_f$. Note that V is foliated by stable curves.

After shrinking the neighborhood $\mathcal{N}_f$ it is enough to prove the proposition for V: just note that there is n_0 with $f^{n_0}(\mathbb{B}_n) \subset \text{int}(V)$, thus taking $\mathcal{N}_f$ small enough we also have that $f^{n_0}_{\mathbf{t}}(\mathbb{B}_n) \subset \text{int}(V)$ for every sequence $\mathbf{t}$ as in the proposition.

Lemma 2.7. For every sequence $\mathbf{t}$ as in the proposition and every point $x \in \mathcal{F}^c_{\text{loc}}(x_f)$ the intersection $\mathcal{F}^{ss}_{\text{loc}}(x) \cap \Gamma_{\mathbf{t}}$ is either a point or empty.

As the set $\Gamma_{\mathbf{t}}$ is connected, its projection in $\mathcal{F}^c_{\mathrm{loc}}(x_f)$ along the strong stable leaves is a closed curve (maybe degenerate to a point), hence $\Gamma_{\mathbf{t}}$ is a continuous graph over that curve.

Proof of Lemma 2.7. Given a point $x \in \Gamma_{\mathbf{t}}$ consider the sequences of points $x_k = f_{\mathbf{t}}^{-k}(x) \in V$ and of discs

$$C(x_k, f_{\mathrm{t}}^{-k}(\mathcal{F}^{ss}_{\mathrm{loc}}(x)) \cap V) = \Delta^{ss}_{\mathbf{t}}(x_k).$$

By the comments in Section 2.1.2, the set $\Delta^{ss}_{\mathbf{t}}(x_k)$ is a stable disc and there is a constant $C > 0$ independent of x such that

$$\operatorname{diam}(\Delta_k(x)) \le C\lambda^k, \quad \text{where} \quad \Delta_k(x) = f_{\mathrm{t}}^k(\Delta^{ss}_{\mathbf{t}}(x_k)) \subset \mathcal{F}^{ss}_{\mathrm{loc}}(x).$$

This immediately implies that the intersection $\mathcal{F}^{ss}_{\mathrm{loc}}(x) \cap \Gamma_{\mathbf{t}}$ consists of at most one point, proving the lemma. □

If $\Gamma_{\mathbf{t}}$ is a single point, we are done. Otherwise, since $\Gamma_{\mathbf{t}}$ is a continuous graph over a curve in $\mathcal{F}^c_{\mathrm{loc}}(x_f)$, thus $\Gamma_{\mathbf{t}}$ is a closed curve. So, it remains to see that this curve is C^1. To see why this is so, denote by $C^2(\mathcal{F}^c_{\mathrm{loc}}(x_f), V)$ the space of C^2 maps from $\mathcal{F}^c_{\mathrm{loc}}(x_f)$ to V. Let $H\colon \mathcal{F}^{ss}_{\mathrm{loc}}(x_f) \to C^2(\mathcal{F}^c_{\mathrm{loc}}(x_f), V)$ be a continuous map which associates to each $x \in \mathcal{F}^{ss}_{\mathrm{loc}}(x_f)$ a C^2 arc $N_x := H(x)(\mathcal{F}^c_{\mathrm{loc}}(x_f))$ such that $H(x_f)$ is the identity and hence $N_{x_f} = \mathcal{F}^c_{\mathrm{loc}}(x_f)$. Thus, by shrinking the neighborhood V if necessary, we can assume that the curve N_x is tangent to the center cone field $\mathcal{C}^c_\beta$ for every $x \in \mathcal{F}^{ss}_{\mathrm{loc}}(x_f)$. Consider now the closed subset

$$\widetilde{V} := \bigcup_{x \in \mathcal{F}^{ss}_{\mathrm{loc}}(x_f)} N_x \subset V. \tag{2.12}$$

So, we may assume that

$$\Gamma_{\mathbf{t}} = \bigcap_{k \ge 0} f_{\mathbf{t}}^k(\widetilde{V}). \tag{2.13}$$

Using Proposition 2.3, the tangent bundle of the random iterates $f_{\mathrm{t}}^k(N_x)$ are Hölder continuous with uniform Hölder constants depending only on $\kappa(\mathcal{F}^c_{loc}(x_f))$. Moreover, by continuity of H and compactness of N_x and $\mathcal{F}^c_{\mathrm{loc}}(x_f)$, $\kappa(N_x)$ has a uniform bound independent of x. Also, by item (c) of Proposition 2.3,

$$J^k_{\mathrm{t},N_x}\colon f_{\mathrm{t}}^k(N_x) \ni y \mapsto \log|\det(Df|_{T_y f_{\mathrm{t}}^k(N_x)})| \tag{2.14}$$

is $(L, 1)$-Hölder continuous, for some $L > 0$ that does not depend on k and x. By (2.13) and (2.14), since $\Gamma_{\mathbf{t}}$ is a closed curve, and using the graph transform approach, $\Gamma_{\mathbf{t}}$ is a C^1 curve. □

Definition 2.8. Let $f \in \mathbb{WC}^2(\mathbb{B}_n)$ and $\mathcal{N}_f$ be the neighborhood of f that satisfies the conclusion of Proposition 2.6. Then, we say the $\mathcal{N}_f$ is a normally hyperbolic adapted C^2 neighborhood of f.

Remark 2.9. Let $f \in \mathbb{WC}^2(\mathbb{B}_n)$ and take the neighborhood $\mathcal{N}_f$ of f as in Proposition 2.6. Then for every $g \in \mathcal{N}_f$ the following trichotomy holds: g is either a weakly contractive map having a generic parabolic attracting fixed point x_g or a uniformly contracting map or has a normally contracting one dimensional submanifold $\Gamma_g \subset \mathbb{B}_n$.

Theorem 2.10. Let $f \in \mathbb{WC}^2(\mathbb{B}_n)$ and $\mathcal{N}_f$ be a normally hyperbolic adapted C^2 neighborhood of f as in Proposition 2.6. Then for every sequence $\mathbf{t} = (\cdots, t_{-n}, \cdots, t_{-1}, t_0)$ with values in I, such that for each n, $f_{t_{-n}} \in \mathcal{N}_f$ or $f_{t_{-n}}$ is a C^2 injective map with $f_{t_{-n}}(\mathbb{B}_n) \subset \mathrm{int}(\mathbb{B}_n)$, the intersection

$$\Gamma_{\mathbf{t}} = \bigcap_{k\geq 0} \mathbb{B}_{\mathbf{t}}(k), \qquad \mathbb{B}_{\mathbf{t}}(k) := f^k_{\mathbf{t}}(\mathbb{B}_n)$$

is either a single point or a one dimensional C^1 closed submanifold.

Proof. Take the neighborhood V of x_f given by (2.11) and the normally hyperbolic adapted C^2 neighborhood $\mathcal{N}_f$ of f, and let

$$\mathcal{A} = \{n \in \mathbb{N} : f^n_{\mathbf{t}}(\mathbb{B}_n) \subset \mathsf{int}(V)\}.$$

There are two cases:

1) $\mathcal{A}$ is finite. Let n_0 be the maximum of $\mathcal{A}$. Then, for each $n > n_0$, $f^n_{\mathbf{t}}(\mathbb{B}_n)$ intersects the complement $\mathbb{B}_n \setminus V$. Thus, by construction, $f_{\mathrm{t}_{-n-1}}$ is uniformly contracting on $f^n_{\mathbf{t}}(\mathbb{B}_n)$ and hence, $\Gamma_{\mathbf{t}}$ is a single point.

2) $\mathcal{A}$ is infinite. Since $\mathcal{A}$ is countable, we can write the elements of $\mathcal{A}$ as a sequence $\{n_k : k \in \mathbb{N}\}$. Thus, for each k, $f^{n_k}_{\mathbf{t}}(\mathbb{B}_n) \subset \mathsf{int}(V)$. We take the sequence $\mathbf{s} = (\cdots, s_{-k}, \cdots, s_{-1}, s_0)$ with values in I and define $g_{s_{-k}}$ by $g_{s_0} = f^{n_1}_{\mathbf{t}}$, and $g_{s_{-k}} = f_{t_{-n_k-1}} \circ \cdots \circ f_{t_{-n_{k+1}}}$, for each $k \geq 1$. Then, by definition, we can apply the argument used in Proposition 2.6 to the maps $g_{s_{-k}}$, $k \geq 0$, we find that the set $\Gamma_{\mathbf{t}}$ is equal to $\bigcap_{k\geq 0} \mathbb{B}_{\mathbf{s}}(k)$, where $\mathbb{B}_{\mathbf{s}}(k) = g^k_{\mathbf{s}}(\mathbb{B}_n)$. Thus, $\bigcap_{k\geq 0} \mathbb{B}_{\mathbf{s}}(k)$ is either a single point or a one dimensional C^1 closed submanifold and the same holds for $\Gamma_{\mathbf{t}}$. This finishes the proof. □

3. Weakly blending regions

We now introduce two local tools used to construct porcupine-like attractors: the covering property and the weak blending region.

Definition 3.1 (Covering property [29]). A finite collection $\mathcal{F} = \{f_1, \dots, f_k\}$ of continuous maps $f_i \colon \mathbb{B}_n \to \mathbb{B}_n$ satisfies the covering property with respect to an open ball $B \subset \mathsf{int}(\mathbb{B}_n)$ if

$$\mathsf{clos}(B) \subset f_1(B) \cup \ldots \cup f_k(B).$$

In this case, we say that the set B is a blending region of $\mathcal{F}$. This blending region is uniformly contracting if the restriction of f_i to $\mathsf{clos}(B)$ is uniformly contracting for every $i = 1, \cdots, k$.

In what follows, we denote by $\mathsf{Inj}^2(\mathbb{B}_n)$ the set of C^2 injective endomorphisms f of $\mathbb{B}_n$ satisfying $f(\mathbb{B}_n) \subset \mathrm{int}(\mathbb{B}_n)$.

Definition 3.2 (Weakly attracting pair). Consider an open set $B \subset \mathsf{int}(\mathbb{B}_n)$ and a finite family of maps $\mathcal{F} = \{f_1, \dots, f_k\}$, $k \geq 3$, with $f_i \in \mathsf{Inj}^2(\mathbb{B}_n)$. We say that $(B, \mathcal{F})$ is a weakly attracting pair if the following conditions hold:

(H1) $f_1 \in \mathbb{WC}^2(\mathbb{B}_n)$ with parabolic fixed point x_{f_1}.

(H2) For $i \geq 2$, the map f_i is uniformly contracting. In particular, f_i has a unique fixed point denoted by x_{f_i}.

(H3) The following contraction on average property holds:

$$\sum_{i=1}^{k} \log \|Df_i(x)\| < 0, \tag{3.1}$$

for every $x \in \mathbb{B}_n$.

(H4) The open set B is a (uniformly contracting) blending region for $\{f_2, \dots, f_k\}$ satisfying

$$\mathsf{clos}(B) \cap \{x_{f_1}, x_{f_2}, \dots, x_{f_k}\} = \emptyset.$$

If $(B, \mathcal{F})$ is a weakly attracting pair then we say that B is a weakly blending region of $\mathcal{F}$.

A pair $(B, \mathcal{G})$ is weakly repelling if $(B, \mathcal{G}^{-1})$ is weakly attracting, where $\mathcal{G}^{-1} = \{g^{-1} \colon g \in \mathcal{G}\}$.

Theorem 3.3. For every $n \geq 1$, there are an open ball B with $\mathsf{clos}(B) \subset \mathsf{int}(\mathbb{B}_n)$ and a finite family of maps $\mathcal{F}$ such that $(B, \mathcal{F})$ is a weakly attracting pair.

Proof. Let $\mathbb{B}_n$ be the closed unit ball containing the origin. Take a point $x_0 \in \mathbb{B}_n$ on the x_n-axis and a smooth function $\alpha : \mathbb{B}_n \to [1/2, 1]$ such that $\alpha(x_0) = 1$, $\alpha(x) \neq 1$ for all $x \in \mathbb{B}_n \setminus \{x_0\}$, and $\alpha(x) = 1/2$ if $\|x\| \geq 1/2$. Let $A(x)$, $x \in \mathbb{B}_n$, be a diagonal matrix with

$A(x) = \mathrm{diag}(\lambda_1, \dots, \lambda_{n-1}, \alpha(x))$, where $0 < \lambda_i < 1/3$ and $\lambda_i \neq \lambda_j$, for every $i, j \in \{1, \dots, n-1\}$ with $i \neq j$. Take the diffeomorphism $\varphi\colon \mathbb{R}^n \to \mathbb{R}^n$ such that

$$\varphi(x_0) = x_0, \quad \text{and} \quad D\varphi(x) = A(x).$$

Then φ has a generic parabolic attracting fixed point at x_0 whose one dimensional central bundle E^c is the x_n-axis. Moreover, if x_0 is close enough to the origin the restriction map $f_1 = \varphi_{|\mathbb{B}_n}$ belongs to $\mathbb{WC}^2(\mathbb{B}_n)$.

Associated to f_1 we consider a region W_1 as in (2.8) and its associated cusp-like region $W = \bigcup_{k=1}^{\infty} W_k$, where $W_k = f_1(W_{k-1})$ for $k \geq 2$, as in (2.9). Taking x_0 close enough to the origin and choosing W on the appropriate side of x_0 (in the central leaf), we can assume that $\mathrm{int}(W)$ contains the origin. Note that $x_0 \notin \mathrm{int}(W)$. Moreover, the map f_1 and the region W satisfy the conclusion of Proposition 2.5.

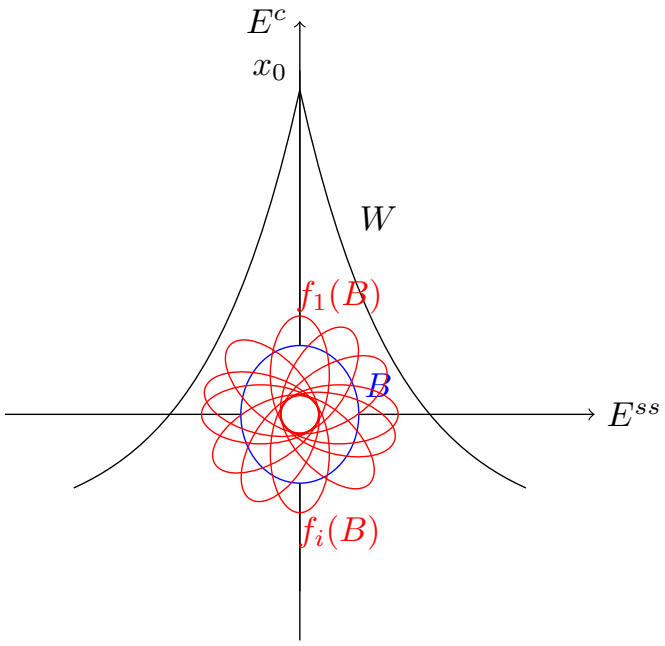


Figure 3. The covering property: The cusp-like region W and the parabolic fixed point x_0 of f. The blending region B and their images $f_i(B)$.

Consider now the open ball $B = B(0, \delta)$, small $\delta > 0$ with $B(0, 5\delta) \subset \mathrm{int}(W)$. Hence $x_0 \notin B$. See Figure 3. Consider now a contraction $g \in \mathrm{Inj}^2(\mathbb{B}_n)$ whose restriction to the ball $B(0, 4\,\delta)$ is given by

$$g(x_1, \dots, x_n) = \frac{9}{10}(x_1, \dots, x_n) + (0, \dots, 0, \delta/5).$$

Then $p_\delta = (0, \dots, 0, 2\,\delta) \notin B$ is the fixed point of g, the origin belongs to $g(B) \cap B$ (note that $g(0, \dots, 0, -10\,\delta/45) = (0, \dots, 0)$), and a neighbourhood of the segment $\{0^{n-1}\} \times [0, \delta]$

is contained in $g(B) \cap B$. This implies that there is a family of linear isometries $A_2, \dots, A_k$ which fixes 0 such that

$$B \subset \bigcup_{i=2}^{k} A_i(g(B)). \tag{3.2}$$

By [12], if A is an isometry of $\mathbb{R}^n$ that fixes 0 then there is an orthogonal decomposition $\mathbb{R}^n = W_1 \bigoplus W_2 \bigoplus \cdots \bigoplus W_m$ such that $dim(W_i) = 1$ or 2 for all i, and the restriction of A to W_i is a rotation unless $i = m$, $dim(W_m) = 1$, and $\det A = -1$, in which case the restriction of A to W_m is a reflection. For each i let

$$f_i = A_i \circ g \circ A_i^{-1}.$$

The above fact implies that the map $f_i \in \mathrm{Inj}^2(\mathbb{B}_n)$ is a contraction whose fixed point $A_i(p_\delta)$ has modulus 2δ and hence does not belong to B. In this way we obtain (H1)–(H3).

To complete (H4), note that $B = A_i(B)$, hence

$$f_i(B) = A_i \circ g \circ A_i^{-1}(B) = A_i \circ g \circ A_i^{-1}(A_i(B)) = A_i(g(B)).$$

Thus, by (3.2),

$$B \subset \bigcup_{i=2}^{k} f_i(B).$$

This implies that B is a blending region disjoint from the fixed points of $f_1, \dots, f_k$, proving the theorem. □

According to the above arguments, we will have the following corollary.

Corollary 3.4. Consider a weakly attracting pair $(B, \mathcal{F})$, where $\mathcal{F} = \{f_1, \dots, f_k\}$. Then there exist C^2 neighbourhoods $\mathcal{N}_i$ of f_i, $i = 1, \dots, k$, such that for every family $\mathcal{G} = \{g_1, \dots, g_k\}$, with $g_i \in \mathcal{N}_i$, the pair $(B, \mathcal{G})$ satisfies conditions (H2)-(H4) in Definition 3.2. Moreover, by Proposition 2.6, g_1 is either a weakly contractive map or a uniformly contracting map or it is normally contracting along some one dimensional submanifold of $\mathbb{B}_n$.

4. Skew products and geometrical structures of their attracting invariant graphs

4.1. Invariant graphs and maximal attractors. As mentioned earlier, we now focus on skew products having non-uniform contraction rates along the fibers. In what follows, we consider compact metric spaces B and Y and continuous maps $h\colon B \to B$. Fixed h, as before, we denote by $C_h(B \times Y)$ the set of skew products

$$F = F_{(h,f_b)}\colon B \times Y \to B \times Y, \qquad F(b, y) = (h(b), f_b(y)), \tag{4.1}$$

whose fiber maps f_b, $b \in B$, are Lipschitz and continuously depend on b. In what follows, we let

$$\Delta(Y) := \{(y, y') \in Y \times Y \colon y \neq y'\}.$$

Definition 4.1 (Contracting on average, [33]). Consider a skew product $F_{(h,f_b)}$ as above and a h-invariant measure m. We say that F is m-contracting on average if the limit

$$\lambda(b) := \lim_{n\to\infty} \frac{1}{n} \sup_{(y,y')\in\Delta(Y)} \log \frac{d(f_b^n(y), f_b^n(y'))}{d(y, y')}, \tag{4.2}$$

exists for m-almost every b, the map $b \mapsto \lambda(b)$ is measurable, and there is $\lambda < 0$ such that

$$\lambda(b) \leq \lambda, \quad \text{for } m\text{-almost every } b. \tag{4.3}$$

In our context, the importance of the contracting on average property is illustrated by the following theorem.

Theorem 4.2. ([33, Theorem 1.4]) Consider compact metric spaces B and Y, a homeomorphism $h \colon B \to B$, a h-invariant ergodic measure m, and an m-contracting on average skew product $F \in C_{h,}(B \times Y)$. Then there exist a h-invariant set $\Lambda \subset B$ with $m(\Lambda) = 1$ and a function $\gamma \colon \Lambda \to Y$ whose graph is invariant and attracting under F.

Denote by

$$C_h^2(B \times Y) \subset C_{h,}(B \times Y) \tag{4.4}$$

the set of skew products whose fiber maps f_b, $b \in B$, are C^2 and continuously depend on b with respect to C^2 topology.

Lemma 4.3. Let $F = F_{(h,f_b)} \in C_h^2(B \times Y)$ be a m-contracting on average skew product with respect to a h-invariant ergodic measure m. Then there exists an open neighborhood $\mathcal{V}(F) \subset C_h^2(B \times Y)$ of F consisting of m-contracting on average skew products.

Proof. Since the map $(b, y) \mapsto f_b(y)$ is continuous and C^2 in y, there exists $\varsigma \in L^1(B, m)$ such that

$$d(f_b(y), f_b(y')) < \varsigma(b)\, d(y, y'),$$

for m-almost every b. Then, by [33, Lem. 5.1], for every $n \geq 0$ the number

$$\varsigma_{F,n}(b) \stackrel{\text{def}}{=} \varsigma_n(b) \stackrel{\text{def}}{=} \sup_{(y,y')\in\Delta(Y)} \frac{d(f_b^n(y), f_b^n(y'))}{d(y, y')} \tag{4.5}$$

exists for almost every b and the map ς_n is measurable. Also, by [33, Lem. 5.3], the sequence ς_n is submultiplicative and hence $\log \varsigma_n$ is subadditive, that is,

$$\log \varsigma_{n+m}(b) \le \log \varsigma_n(b) + \log \varsigma_m(h^n(b)).$$

We can thus apply Kingman's subadditive ergodic theorem [23, Thm. 5.4] to $\log \varsigma_n$, obtaining that the map

$$\lambda(b) = \lim_{n\to+\infty} \frac{1}{n} \log \varsigma_n(b) \tag{4.6}$$

exists for m-almost every b and is h invariant. Moreover, by subadditivity,

$$\lim_{n\to\infty} \frac{1}{n} \log \varsigma_n(b) = \inf_n \frac{1}{n} \log \varsigma_n(b) \tag{4.7}$$

and, by ergodicity of m, it is constant m-a.e., say equal to λ_F. Since F is contracting on average we have that $\lambda_F < 0$. Fix small $\delta > 0$ and take any $G = G_{(h,g_b)} \in C^2_h(B \times Y)$ that is δ-close to F with respect to the metric in (4.12), that is, $\sup_b \operatorname{dist}_{C^2}(f_b^{\pm}, g_b^{\pm}) < \delta$. We claim that, for small δ, G is contracting on average with respect to m. Indeed, as above, the sequence

$$\varsigma_{G,n}(b) \stackrel{\text{def}}{=} \sup_{(y,y'\in\Delta(Y))} \frac{d(g_b^n(y), g_b^n(y'))}{d(y,y')}. \tag{4.8}$$

is subadditive and hence both statements (4.6) and (4.7) hold for $\log \varsigma_{G,n}$ (where in (4.6) we consider $\lambda_G(b)$). In particular, by the ergodicity of m,

$$\lim_{n\to+\infty} \frac{1}{n} \log \varsigma_{G,n}(b) = \inf_n \frac{1}{n} \log \varsigma_{G,n}(b)$$

is constant m-a.e., say equal to λ_G. To prove the lemma, we need to see that $\lambda_G < 0$. Fix small $\varepsilon > 0$ with $\lambda_F + 2\varepsilon < 0$. By (4.7), $\lambda_F = \inf_n \frac{1}{n} \log \varsigma_n(b)$, for m-a.e. b. Thus, for a typical point b for λ_F and λ_G, there exists $n_0 = n_0(b) \in \mathbb{N}$ such that

$$\frac{1}{n_0} \log \varsigma_{F,n_0}(b) < \lambda_F + \varepsilon.$$

Taking δ small enough, we get

$$\frac{1}{n_0} \log \varsigma_{G,n_0}(b) < \lambda_F + 2\varepsilon < 0.$$

Consequently,

$$\lambda_G = \inf_n \frac{1}{n} \log \varsigma_{G,n} < \frac{1}{n_0} \log \varsigma_{G,n_0}(b) < \lambda_F + 2\varepsilon < 0,$$

proving that G is contracting on average with respect to m. □

In this section, we introduce a class of skew products driven by an expanding circle map and their solenoidal extensions. We show that maximal attractors of the extension skew products are either an attracting continuous invariant graph or a thick n-porcupine graph.

4.2. Skew products driven by expanding circle maps and their extensions. Given $\ell \in \mathbb{N}$, $\ell \geq 2$, we consider the expanding map of the circle

$$\varphi_\ell \colon \mathbb{T}^1 \to \mathbb{T}^1, \quad \varphi_\ell(t) = \ell t \pmod 1.$$

Take skew product maps defined on $\mathbb{T}^1 \times \mathbb{B}_n$ driven by φ_ℓ of the form

$$F_{(\ell, f_t)} \colon \mathbb{T}^1 \times \mathbb{B}_n \to \mathbb{T}^1 \times \mathbb{B}_n, \quad F(t,x) = (\varphi_\ell(t), f_t(x)), \tag{4.9}$$

where the fiber maps $f_t \colon \mathbb{B}_n \to \mathbb{B}_n$, $t \in \mathbb{T}^1$, are C^2 diffeomorphisms from the compact ball $\mathbb{B}_n$ onto their images such that $f_t(\mathbb{B}_n) \subset \mathrm{int}(\mathbb{B}_n)$. Since the maps φ_ℓ are not invertible, we consider their extensions given by the inverse limit construction in [36]. For that consider the set

$$\mathcal{S}_\ell := \{(\ldots, t_{-1}, t_0) \in (\mathbb{T}^1)^{\mathbb{Z}^-} : t_{-j} = \varphi_\ell(t_{-j-1})\}$$

endowed with the product topology. The extension ξ_ℓ of φ_ℓ is defined by

$$\xi_\ell \colon \mathcal{S}_\ell \to \mathcal{S}_\ell, \quad \xi(\mathrm{t}) = (\ldots, t_{-1}, t_0, \varphi_\ell(t_0)), \text{ for } \mathrm{t} = (\ldots, t_{-1}, t_0).$$

The extension map ξ_ℓ is invertible and its inverse is given by

$$\xi_\ell^{-1}(\mathrm{t}) = (\ldots, t_{-2}, t_{-1}).$$

Finally, consider the projection

$$\pi_\ell \colon \mathcal{S}_\ell \to \mathbb{T}^1, \quad \pi(\mathrm{t}) = t_0. \tag{4.10}$$

Note that $\pi_\ell \circ \xi_\ell = \varphi_\ell \circ \pi_\ell$, which means that π_ℓ is a semiconjugacy between φ_ℓ and ξ_ℓ. The skew product map $F_{(\ell, f_t)}$ in (4.9) induces the skew product map $\mathbb{F}_F = \mathbb{F}_{(\xi_\ell, f_\mathrm{t})}$ on $\mathcal{S}_\ell \times \mathbb{B}_n$, called the extension of F, defined by

$$\mathbb{F}_F(\mathrm{t}, x) = (\xi_\ell(\mathrm{t}), f(\mathrm{t}, x)) = (\xi_\ell(\mathrm{t}), f_\mathrm{t}(x)), \tag{4.11}$$

where $f_\mathrm{t}(x) := f_{t_0}(x)$.

Remark 4.4. The fiber maps of the skew product maps $F_{(\ell, f_t)}$ and $\mathbb{F}_F$ are the same.

Remark 4.5. On the circle $\mathbb{T}^1$ and the inverse limit spaces $\mathcal{S}_\ell$, we use Borel σ-algebras. Note that the expanding map φ_ℓ possesses an absolutely continuous invariant ergodic measure ν_ℓ^+ whose density is bounded away from zero and is equivalent to Lebesgue [24, 27]. The extension map ξ_ℓ has the invariant measure ν_ℓ defined on $\mathcal{S}_\ell$ inherited from ν_ℓ^+ defined by

$$\nu_\ell(\{t_{-n} \in I_n, \cdots, t_0 \in I_0\}) := \nu_\ell^+(\varphi_\ell^{-n}(I_0) \cap \varphi_\ell^{-n+1}(I_1) \cap \cdots \cap I_n),$$

where I_i, $i = 0, \cdots, n$, are intervals of $\mathbb{T}^1$.

Notations 4.6. The following notations are introduced:

i) The space of all skew products $F_{(\ell,f_t)}$ as in (4.9) is denoted by $C_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$. We denote by $C(\mathcal{S}_\ell \times \mathbb{B}_n)$ the space of skew product maps $\mathbb{F} = \mathbb{F}_F$ which are an extension of some $F \in C_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$. We equip these spaces with the metric given by (4.12)

ii) Given $F_{(\ell,f_t)}$, the set of all F-invariant probability measures on $\mathbb{T}^1 \times \mathbb{B}_n$ with marginal ν_ℓ^+ is denoted by $\mathcal{M}_{\nu_\ell^+}(F)$.

iii) Similarly, for $\mathbb{F} = \mathbb{F}_F$, we denote by $\mathcal{M}_{\nu_\ell}(\mathbb{F})$ the set of all $\mathbb{F}$-invariant probability measures on $\mathcal{S}_\ell \times \mathbb{B}_n$ with marginal ν_ℓ.

We equip the space $C_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$ with the following metric:

$$\operatorname{dist}_{C_\ell}(F,G) := \sup_{t\in\mathbb{T}^1}(\operatorname{dist}_{C^2}(f_t^{\pm 1}, g_t^{\pm 1})), \text{ for each } F,G \in C_\ell(\mathbb{T}^1 \times \mathbb{B}_n). \tag{4.12}$$

The following lemma [13] relates invariant measures for a skew product and its extension.

Lemma 4.7 ([13] and Lemma 2.1 in [20]). Let $F_{(\ell,f_t)}$ and consider any invariant measure $\mu^+ \in \mathcal{M}_{\nu_\ell^+}(F)$ with disintegration $(\mu_t^+)_{t\in\mathbb{T}^1}$. Then there is an invariant measure $\mu \in \mathcal{M}_{\nu_\ell}(\mathbb{F}_F)$ whose disintegration $(\mu_t)_{t\in\mathcal{S}_\ell}$ is given by

$$\mu_t = \lim_{n\to\infty} f_{t_{-n}}^n \mu_{t_{-n}}^+$$

for ν_ℓ-almost every $(\ldots, t_{-1}, t_0) \in \mathcal{S}_\ell$.

4.3. The subset $\mathcal{A}_{\ell,\mathbb{T}^1} = \mathcal{A}_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$ of $C_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$.

In what follows, we define the subset $\mathcal{A}_{\ell,\mathbb{T}^1}$.

Definition 4.8 (The sets $\mathcal{A}_{\ell,\mathbb{T}^1}$ and $\mathcal{A}_{\mathcal{S}_\ell}$). Given $\ell \geq 2$, denote by $\mathcal{A}_{\ell,\mathbb{T}^1} = \mathcal{A}_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$ the subset of $C_\ell(\mathbb{T}^1 \times \mathbb{B}_n)$ consisting of all skew product maps $F_{(\varphi_\ell,f_t)}$ which are contracting on average with respect to the measure ν_ℓ^+ and such that there are an open ball $B \subset \operatorname{int}(\mathbb{B}_n)$, disjoint closed intervals $\{L_1, \ldots, L_k\}$ of $\mathbb{T}^1$, with $\ell/2 \geq k \geq 3$, and a family $\mathcal{F} = \{f_{t_1}, \ldots, f_{t_k}\}$ of fiber maps of F with $t_i \in L_i$ for every $i = 1, \ldots, k$ such that:

(A1) $(B, \mathcal{F})$ is a weak attracting pair.

(A2) (i) $f_1 \in \mathbb{WC}^2(\mathbb{B}_n)$ and there is a normally hyperbolic adapted C^2 neighborhood $\mathcal{N}_{f_1}$ of f_1 (see Definition 2.8) such that $f_t \in \mathcal{N}_{f_1}$ for every $t \in L_1$,

(ii) for every $s_i \in L_i$, $i = 2, \ldots, k$, the fiber maps $\{f_{s_2}, \ldots, f_{s_k}\}$ satisfy the covering property

$$\operatorname{clos}(B) \subset f_{s_2}(B) \cup \cdots \cup f_{s_k}(B);$$

(iii) for every $t \notin L_1$ the fiber map f_t of F is uniformly contracting.

(A3) $\varphi_\ell(L_i) = \mathbb{T}^1$ for every $i = 1, \dots, k$.

We define the subset $\mathcal{A}_{\mathcal{S}_\ell} = \mathcal{A}(\mathcal{S}_\ell \times \mathbb{B}_n)$ of $\mathcal{C}(\mathcal{S}_\ell \times \mathbb{B}_n)$ as follows:

$$\mathcal{A}_{\mathcal{S}_\ell} \stackrel{\text{def}}{=} \left\{ \mathbb{F} = \mathbb{F}_F \in \mathcal{C}(\mathcal{S}_\ell \times \mathbb{B}_n) \ , \text{such that } F \in \mathcal{A}_{\ell,\mathbb{T}^1} \right\}.$$

In Section 5.5, we will show that the sets $\mathcal{A}_{\ell,\mathbb{T}^1}$ (and hence the sets $\mathcal{A}_{\mathcal{S}_\ell}$) are not empty for every ℓ large enough.

Theorem 4.9. For every ℓ sufficiently large, there is an open neighbourhood $\mathcal{V}_{\mathcal{S}_\ell}$ of $\mathcal{A}_{\mathcal{S}_\ell}$ in $\mathcal{C}(\mathcal{S}_\ell \times \mathbb{B}_n)$ such that every $\mathbb{F} \in \mathcal{V}_{\mathcal{S}_\ell}$ has a maximal attractor $A_{max}(\mathbb{F})$ that is topologically transitive and satisfies the following dichotomy:

- either $A_{max}(\mathbb{F})$ is an attracting continuous invariant graph;
- or $A_{max}(\mathbb{F})$ is a solenoidal n-porcupine-like graph.

The proof of this theorem is given in Subsection 4.4. By Lemma 4.3, to be contracting on average is an open condition. So, fixed ℓ, we can take an open set $\mathcal{W}_{\ell,\mathbb{T}^1} \supset \mathcal{A}_{\ell,\mathbb{T}^1}$ consisting of skew product maps $F_{(\varphi_\ell, f_t)}$ which are contracting on average with respect to ν_ℓ^+. We let

$$\mathcal{W}_{\mathcal{S}_\ell} \stackrel{\text{def}}{=} \left\{ \mathbb{F} = \mathbb{F}_F \in \mathcal{C}(\mathcal{S}_\ell \times \mathbb{B}_n) \ , \text{such that } F \in \mathcal{W}_{\ell,\mathbb{T}^1} \right\}. \tag{4.13}$$

Remark 4.10. Since π_ℓ, given by (4.10), is a semiconjugacy between φ_ℓ and ξ_ℓ, and by Remark 4.4, the fiber maps of F and $\mathbb{F} = \mathbb{F}_F$ are the same, thus, $\mathcal{W}_{\mathcal{S}_\ell}$ is open in $\mathcal{C}(\mathcal{S}_\ell \times \mathbb{B}_n)$.

Consider $\mathbb{F} = \mathbb{F}_{(\xi_\ell, \mathrm{f_t})} \in \mathcal{W}_{\mathcal{S}_\ell}$. By definition, $\mathbb{F}$ is contracting on average with respect to ν_ℓ. Since $\mathcal{S}_\ell$ and $\mathbb{B}_n$ are compact metric spaces, ξ_ℓ is a homeomorphism, ν_ℓ is an invariant ergodic measure, and the fiber maps f_t are C^2 diffeomorphisms onto their images, we can apply Theorem 4.2 to $\mathbb{F}$, getting the next result.

Proposition 4.11. Let $\mathbb{F} \in \mathcal{W}_{\mathcal{S}_\ell}$. Then there exist a ξ_ℓ-invariant set $\widetilde{\mathcal{S}}_\ell \subseteq \mathcal{S}_\ell$ with $\nu_\ell(\widetilde{\mathcal{S}}_\ell) = 1$ and a measurable function $\gamma_\mathbb{F} \colon \widetilde{\mathcal{S}}_\ell \to \mathbb{B}_n$ whose graph is invariant and attracting under $\mathbb{F}$.

4.4. Skew products with high dimensional porcupine graph. By shrinking the set $\mathcal{W}_{\ell,\mathbb{T}^1}$, we may assume that every fiber map f_t of any $F \in \mathcal{W}_{\ell,\mathbb{T}^1}$ belongs to $\mathrm{Inj}^2(\mathbb{B}_n)$. By Remark

4.4, the same holds for any $\mathbb{F} = \mathbb{F}_F \in \mathcal{W}_{\mathcal{S}_\ell}$. This implies that $\mathcal{S}_\ell \times \mathbb{B}_n$ is a trapping region of $\mathbb{F}$, hence its maximal attractor

$$A_{max}(\mathbb{F}) \stackrel{\text{def}}{=} \bigcap_{n\geq 0} \mathbb{F}^n(\mathcal{S}_\ell \times \mathbb{B}_n) \tag{4.14}$$

is well defined. In what follows, the integer ℓ remains fixed and hence the subscript ℓ will be omitted.

Proof of Theorem 4.9. Take $\mathbb{F} = \mathbb{F}_F \in \mathcal{W}_{\mathcal{S}_\ell}$. By Proposition 4.11, there are a ξ-invariant set $\widetilde{\mathcal{S}} \subset \mathcal{S}$ with $\nu(\widetilde{\mathcal{S}}) = 1$ and a measurable function $\gamma_{\mathbb{F}} : \widetilde{\mathcal{S}} \to \mathbb{B}_n$ whose graph $\gamma_{\mathbb{F}}$, denoted by $\Gamma_{\mathbb{F}}$, is $\mathbb{F}$-invariant.

Lemma 4.12. Let $\mathbb{F} = \mathbb{F}_F \in \mathcal{W}_{\mathcal{S}_\ell}$. Then $\Gamma_{\mathbb{F}} \subset A_{max}(\mathbb{F})$. Moreover, if $\widetilde{\mathcal{S}} = \mathcal{S}$ then $\Gamma_{\mathbb{F}} = A_{max}(\mathbb{F})$.

Proof. For each $\mathrm{t} = (\ldots, t_{-n} \ldots, t_{-1}, t_0) \in \mathcal{S}$ define

$$\mathbb{B}^k(\mathrm{t}, \mathbb{F}) \stackrel{\text{def}}{=} f_{\xi_\ell^{-1}(\mathrm{t})} \circ \cdots \circ f_{\xi_\ell^{-k}(\mathrm{t})}(\mathbb{B}_n) = f_{t_{-1}} \circ \cdots \circ f_{t_{-k}}(\mathbb{B}_n). \tag{4.15}$$

As the fiber maps $f_{t_{-j}}$ belong to $\mathrm{Inj}^2(\mathbb{B}_n)$ the sets $\mathbb{B}^k(\mathrm{t}, \mathbb{F})$ are well defined and form a nested sequence of compact sets. Thus,

$$\mathbb{B}^\infty(\mathrm{t}, \mathbb{F}) \stackrel{\text{def}}{=} \bigcap_{k=0}^{\infty} \mathbb{B}^k(\mathrm{t}, \mathbb{F}) \neq \emptyset. \tag{4.16}$$

Note that

$$A_{max}(\mathbb{F}) \bigcap (\{\mathrm{t}\} \times \mathbb{B}_n) = \mathbb{B}^\infty(\mathrm{t}, \mathbb{F}). \tag{4.17}$$

Note that, $\Gamma_{\mathbb{F}}$ is attracting, so, for each $\mathrm{t} \in \widetilde{\mathcal{S}}$,

$$\lim f_{\xi_\ell^{-1}(\mathrm{t})} \circ \cdots \circ f_{\xi_\ell^{-n}(\mathrm{t})}(\mathbb{B}_n) = \lim f_{\xi_\ell^{-1}(\mathrm{t})} \circ \cdots \circ f_{\xi_\ell^{-n}(\mathrm{t})}(\gamma_{\mathbb{F}}(\mathrm{t})) = B^\infty(\mathrm{t}, \mathbb{F}).$$

Since the graph $\gamma_{\mathbb{F}}$ is invariant, we get

$$\Gamma_{\mathbb{F}} = \bigcup_{\mathrm{t}\in\widetilde{\mathcal{S}}} B^\infty(\mathrm{t}, \mathbb{F}) \subset A_{max}(\mathbb{F}),$$

proving the first part of the lemma. For the second part, assuming that $\widetilde{\mathcal{S}} = \mathcal{S}$, note that by (4.17) it holds

$$A_{max}(\mathbb{F}) = \bigcup_{\mathrm{t}\in\mathcal{S}} \Big(A_{max}(\mathbb{F}) \cap \big(\{\mathrm{t}\} \times \mathbb{B}_n\big)\Big) = \bigcup_{\mathrm{t}\in\mathcal{S}} \{\mathrm{t}\} \times \mathbb{B}^\infty(\mathrm{t}, \mathbb{F}) = \Gamma_{\mathbb{F}},$$

which ends the proof of the lemma. □

By Lemma 4.12, if $\widetilde{\mathcal{S}} = \mathcal{S}$ then $A_{max}(\mathbb{F})$ is an invariant graph and the graph function $\gamma_{\mathbb{F}}$ is defined on the whole space $\mathcal{S}$. In this case, we have $\Gamma_{\mathbb{F}} = A_{max}(\mathbb{F})$. For the case that $\widetilde{\mathcal{S}} \neq \mathcal{S}$, the lemma below states the structure of non-trivial spines.

Lemma 4.13. Let $\mathbb{F} \in \mathcal{W}_{\mathcal{S}_\ell}$ and there exist $\mathrm{t} \in \mathcal{S} \setminus \widetilde{\mathcal{S}}$ for that $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is nontrivial. Then the set $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is a one dimensional submanifold tangent to the central bundle.

Proof. Let $\mathbb{F} = \mathbb{F}_F \in \mathcal{W}_{\mathcal{S}_\ell}$ with nontrivial spine $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$, for $\mathrm{t} \in \mathcal{S} \setminus \widetilde{\mathcal{S}}$. Then $F \in \mathcal{W}_{\ell,\mathbb{T}^1}$. By item (*iii*) of Condition ($A2$) of Definition 4.8 and shrinking $\mathcal{W}_{\ell,\mathbb{T}^1}$, for each $t \in \mathbb{T}^1 \setminus L_1$, the fiber maps f_t of F are uniformly contracting. By Remark 4.4, the fiber maps of $\mathbb{F}_F$ and F are the same. So, this fact holds for every $\mathbb{F} \in \mathcal{W}_{\mathcal{S}_\ell}$ and every $\mathrm{t} = (\ldots, t_{-1}, t_0) \in \mathcal{S}$ with $t_0 \in \mathbb{T}^1 \setminus L_1$. Given $\mathrm{t} = (\ldots, t_{-1}, t_0) \in \mathcal{S}$, define

$$n(\mathrm{t}) \stackrel{\text{def}}{=} \sup\{j : t_{-j} \in \mathbb{T}^1 \setminus L_1\}.$$

If $n(\mathrm{t})$ is finite, then for each $j > n(\mathrm{t})$, $t_{-j} \in L_1$. Therefore, By item (*i*) of Condition ($A2$) of Definition 4.8 and Proposition 2.6, the intersection $\bigcap_{m\geq 0} \mathbb{B}^m(\mathrm{t}, \mathbb{F}) = \lim_{n\to\infty} f_{t_{-1}} \circ \cdots \circ f_{t_{-m}}(\mathbb{B}_n)$ is either a single point or a one dimensional C^1 submanifold. The hypothesis that $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is nontrivial, prevents the first case. So, $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is a one dimensional C^1 submanifold. Now, assume that $n(\mathrm{t})$ is infinite. If the set $\{j : t_{-j} \in L_1\}$ is finite, a straightforward argument shows that $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is a single point that contradicts the hypothesis. So, the set $\{j : t_{-j} \in L_1\}$ is infinite. This means that there are infinitely many $t_{-j} \in L_1$ and also infinitely many $t_{-j} \in \mathbb{T}^1 \setminus L_1$. In this case, the proof follows from Theorem 2.10 and the hypothesis that $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is nontrivial. □

Note that

$$A_{max}(\mathbb{F}) = \bigcup_{\mathrm{t}\in\mathcal{S}} \mathbb{B}^\infty(\mathrm{t}, \mathbb{F}).$$

Lemma 4.14. Let $\mathbb{F} \in \mathcal{W}_{\mathcal{S}_\ell}$ and suppose that there exists a nontrivial spine $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$, for some $\mathrm{t} \in \mathcal{S}$. Then the set of nontrivial spines has the cardinality of the continuum and it is dense in $\mathcal{S}$. Furthermore, the same holds for the set $\widetilde{\mathcal{S}}$.

Proof. For $t_j \in \mathbb{T}^1$, $j = 1, \ldots, m$, we denote by $\bar{t}(m) = (t_1, \ldots, t_m)$. Define

$$\mathrm{dist}(\bar{t}(m), \bar{t}'(m)) \stackrel{\text{def}}{=} \max\{d_{\mathbb{T}^1}(t_1, t'_1), \ldots, d_{\mathbb{T}^1}(t_m, t'_m)\},$$

where $\bar{t}(m) = (t_1, \ldots, t_m)$ and $\bar{t}'(m) = (t'_1, \ldots, t'_m)$. Also, given $t_j \in \mathbb{T}^1$, $j = 1, \ldots, m$, denote by $f_{[\bar{t}(m)]} \stackrel{\text{def}}{=} f_{[t_1,\ldots,t_m]} \stackrel{\text{def}}{=} f_{t_1} \circ \cdots \circ f_{t_m}$.

Claim 4.15. For every $m \geq 1$ and $\varepsilon > 0$ there exists $\delta = \delta(m, \varepsilon) > 0$ such that for every $t_j, t'_j \in \mathbb{T}^1$, $j = 1, \ldots, m$, with

$$\mathsf{dist}(\bar{t}(m), \bar{t}'(m)) < \delta$$

then

$$|\mathrm{diam}(f_{[\bar{t}(m)]}(\mathbb{B}_n)) - \mathrm{diam}(f_{[\bar{t}'(m)]}(\mathbb{B}_n))| < \varepsilon.$$

Proof. Fix any $m \in \mathbb{N}$ and $\varepsilon > 0$. Since the map $(\bar{t}(m), y) \mapsto f_{[\bar{t}(m)]}(y)$ is uniformly continuous, thus, there exists $\delta > 0$ such that for each $t_j, t'_j \in \mathbb{T}^1$, $j = 1, \ldots, m$,

$$\max\{\mathsf{dist}(\bar{t}(m), \bar{t}'(m)), d_{\mathbb{B}_n}(x, y)\} < \delta, \quad \text{implies that} \quad d_{\mathbb{B}_n}(f_{[\bar{t}(m)]}(x), f_{[\bar{t}'(m)]}(y)) < \varepsilon/2.$$

Take $\bar{t}(m) = (t_1, \ldots, t_m)$ and $\bar{t}'(m) = (t'_1, \ldots, t'_m)$ in $\mathbb{T}^m$ such that $\mathsf{dist}(\bar{t}(m), \bar{t}'(m)) < \delta$. For $A \subseteq \mathbb{B}_n$, let $U_{\varepsilon/2}(A) \stackrel{\text{def}}{=} \{y \in \mathbb{B}_n : d_{\mathbb{B}_n}(y, A) \leq \varepsilon/2\}$. We need to show that

$$f_{[\bar{t}(m)]}(\mathbb{B}_n) \subset U_{\varepsilon/2}(f_{[\bar{t}'(m)]}(\mathbb{B}_n)) \quad \text{and} \quad f_{[\bar{t}'(m)]}(\mathbb{B}_n) \subset U_{\varepsilon/2}(f_{[\bar{t}(m)]}(\mathbb{B}_n)). \tag{4.18}$$

This implies that

$$\mathrm{diam}(f_{[\bar{t}(m)]}(\mathbb{B}_n)) \leq \mathrm{diam}(f_{[\bar{t}'(m)]}(\mathbb{B}_n)) + \varepsilon \quad \text{and} \quad \mathrm{diam}(f_{[\bar{t}'(m)]}(\mathbb{B}_n)) \leq \mathrm{diam}(f_{[\bar{t}(m)]}(\mathbb{B}_n)) + \varepsilon,$$

proving the claim. We now prove (4.18). Given $y \in f_{[\bar{t}(m)]}(\mathbb{B}_n)$ we can write $y = f_{[\bar{t}(m)]}(x)$ for some $x \in \mathbb{B}_n$. Hence, taking $y' = f_{[\bar{t}'(m)]}(x)$, we have

$$d_{\mathbb{B}_n}(y, y') = d_{\mathbb{B}_n}(f_{[\bar{t}(m)]}(x), f_{[\bar{t}'(m)]}(x)) < \varepsilon/2,$$

proving that $y \in U_{\varepsilon/2}(f_{[\bar{t}'(m)]}(\mathbb{B}_n))$. The proof of the other inclusion of (4.18) is similar. □

In what follows, for each $\mathrm{t} = (\ldots, t_{-m}, \ldots, t_{-1}, t_0) \in \mathcal{S}$ the finite word $\tilde{t}(m) = (t_{-1}, \ldots, t_{-m})$ is called the corresponding m-word of t. Let us take the disjoint closed intervals $I_i \subset \mathbb{T}^1$, $i = 1, \ldots, \ell$, with length $1/\ell$, such that $\mathbb{T}^1 = \bigcup_{i=1}^{\ell} I_i$ and there exist indices i_j, $j = 1, \ldots, k$, so that $I_{i_j} = L_j$, where L_j are disjoint closed intervals of $\mathbb{T}^1$ given by Definition 4.8. Then by the choice of I_i , $\varphi_\ell(I_i) = \mathbb{T}^1$. Fix any $\mathrm{t} = (\ldots, t_{-1}, t_0) \in \mathcal{S}$ for that $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is a non-trivial spine. Take $\varepsilon > 0$ with

$$\varepsilon < \frac{1}{8}\mathrm{diam}(\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})), \tag{4.19}$$

and m sufficiently large such that

$$U_{\varepsilon/2}(f_{[\bar{t}(m)]}(\mathbb{B}_n)) \subset U_\varepsilon(\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})). \tag{4.20}$$

where $\tilde{t}(m) = (t_{-1}, \ldots, t_{-m})$ is the corresponding m-word of t. By Claim 4.15 and inclusions (4.18), for ε given by (4.19) , there exists $\delta > 0$ such that for each $\mathrm{t}' = (\ldots, t'_{-m}, \ldots, t'_{-1}, t'_0) \in \mathcal{S}$ and the corresponding m-word $\bar{t}'(m) = (t'_{-1}, \ldots, t'_{-m})$ of t' , if $\max\{d_{\mathbb{T}^1}(t_0, t'_0), dist(\bar{t}(m), \bar{t}'(m))\} < \delta$ then

$$f_{[\bar{t}(m)]}(\mathbb{B}_n) \subset U_{\varepsilon/2}(f_{[\bar{t}'(m)]}(\mathbb{B}_n)), \quad f_{[\bar{t}'(m)]}(\mathbb{B}_n) \subset U_{\varepsilon/2}(f_{[\bar{t}(m)]}). \tag{4.21}$$

Now, we choose a sequence $\mathrm{t}' = (\dots, t'_{-n} \dots, t'_{-1}, t'_0) \in \mathcal{S}$ with $d_{\mathbb{T}^1}(t_0, t'_0) < \delta$, so that for each j, t_{-j} and t'_{-j} belong to the same closed interval I_{i_j}. Since $\varphi_\ell(t_{-j-1}) = t_{-j}$ and $\varphi_\ell(t'_{-j-1}) = t'_{-j}$, hence $t_{-j-1}, t'_{-j-1} \in \varphi_\ell^{-1}(I_j)$. Note that the inverse branch $\varphi_\ell^{-1}|_{I_j}$ is a contraction with a uniform contracting rate $1/\ell$. Thus, for each j, $d_{\mathbb{T}^1}(t_{-j-1}, t'_{-j-1}) < 1/\ell d_{\mathbb{T}^1}(t_{-j}, t'_{-j})$. Thus, for each j

$$d_{\mathbb{T}^1}(t_{-j}, t'_{-j}) < \delta, \tag{4.22}$$

which implies that $\max\{d_{\mathbb{T}^1}(t_0, t'_0), dist(\bar{t}(m), \bar{t}'(m))\} < \delta$. Then, by (4.20) , (4.21) and (4.22)

$$U_{\varepsilon/2}(f_{[\bar{t}'(m)]}(\mathbb{B}_n)) \subset U_\varepsilon(\mathbb{B}^\infty(\mathrm{t}', \mathbb{F})). \tag{4.23}$$

Therefore, by (4.21) and (4.23)

$$\mathbb{B}^\infty(\mathrm{t}, \mathbb{F}) \subset f_{[\bar{t}(m)]}(\mathbb{B}_n) \subset U_{\varepsilon/2}(f_{[\bar{t}'(m)]}(\mathbb{B}_n)) \subset U_\varepsilon(\mathbb{B}^\infty(\mathrm{t}', \mathbb{F})). \tag{4.24}$$

Consequently, by (4.24) and (4.19), we get

$$\operatorname{diam}(\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})) \leq \operatorname{diam}(\mathbb{B}^\infty(\mathrm{t}', \mathbb{F})) + 2\varepsilon \leq \operatorname{diam}(\mathbb{B}^\infty(\mathrm{t}', \mathbb{F})) + \frac{1}{4}\operatorname{diam}(\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})). \tag{4.25}$$

By (4.25), since $\mathbb{B}^\infty(\mathrm{t}, \mathbb{F})$ is a non-trivial spine, $\mathbb{B}^\infty(\mathrm{t}', \mathbb{F})$ is also a nontrivial spine. As a result, the set of nontrivial spines has the cardinality of the continuum but it is not an open subset of $\mathcal{S}$. Because, for each $t'_0 \in \mathbb{T}^1$ with $d_{\mathbb{T}^1}(t_0, t'_0) < \delta$, there exists only one inverse limit branch $\mathrm{t}' = (\dots, t'_{-m} \dots, t'_{-1}, t'_0)$ that fulfills the construction described above, namely, for each j, t_{-j} and t'_{-j} belong to the same closed interval I_{i_j}. To prove the next statement, take any t' with a nontrivial spine as above such that t'_0 is irrational and let $\mathrm{s}_k = \xi_\ell^k(\mathrm{t}') = (\dots, t'_{-j}, \dots, t'_0, \varphi_\ell(t'_0), \dots, \varphi_\ell^k(t'_0))$, for each positive integer k. Then, by continuity, for each $k \geq 1$, we have

$$\lim_{m\to\infty} f_{[\bar{s}_k(k+m)]}(\mathbb{B}_n) = \lim_{m\to\infty} f_{\varphi_\ell^k(t'_0)} \circ \dots \circ f_{t'_0} \circ \dots \circ f_{t'_{-m}}(\mathbb{B}_n) = f_{\varphi_\ell^k(t'_0)} \circ \dots \circ f_{\varphi_\ell(t'_0)}(\lim_{m\to\infty} f_{[\bar{t}'(m)]}(\mathbb{B}_n)).$$

Consequently, $\mathbb{B}^\infty(\mathrm{s}_k, \mathbb{F})$ is a non-trivial spine. Since t'_0 is irrational, the φ_ℓ-orbit of t'_0 is dense in $\mathbb{T}^1$. Since the base maps φ_ℓ and ξ_ℓ are semiconjugate, the ξ_ℓ-orbit of t' is also dense in $\mathcal{S}$. Thus, the set of nontrivial spines is dense in $\mathcal{S}$. On the other hand, by Proposition 4.11, $\nu_\ell(\widetilde{\mathcal{S}}) = 1$. Also, ν_ℓ inherited from ν_ℓ^+ and ν_ℓ^+ is equivalent to Lebesgue. By these facts, $\widetilde{\mathcal{S}}$ has the cardinality of the continuum and it is dense in $\mathcal{S}$. This finishes the proof. □

By Lemma 4.12, $\Gamma_{\mathbb{F}} \subset A_{max}(\mathbb{F})$, and hence the closure of $\Gamma_{\mathbb{F}}$ is contained in $A_{max}(\mathbb{F})$. The next lemma provides the inclusion in the another direction.

Lemma 4.16. If $\mathbb{F} \in \mathcal{W}_{\mathcal{S}_\ell}$ has invariant graph $\Gamma_{\mathbb{F}}$ and $\mathcal{S} \setminus \widetilde{\mathcal{S}} \neq \emptyset$, then every non-trivial spine is contained in $\overline{\Gamma_{\mathbb{F}}}$ and

$$\overline{\Gamma_{\mathbb{F}}} = A_{\max}(\mathbb{F}).$$

Moreover, $\pi_2(A_{\max}(\mathbb{F}))$ has nonempty interior.

Proof. Recall two facts used below: (i) by Proposition 4.11 there is an attracting invariant graph $\Gamma_{\mathbb{F}} \subset A_{\max}(\mathbb{F})$ defined on $\widetilde{\mathcal{S}}$; (ii) by the covering property in Definition 4.8 there exists an open ball $B \subset \operatorname{int}\mathbb{B}_n$ whose images under appropriate fiber maps cover $\overline{B}$. Fix an arbitrary $(t,x) \in A_{\max}(\mathbb{F})$ and a neighbourhood $U \times V$ of (t,x). Since $\varphi_\ell : \mathbb{T}^1 \to \mathbb{T}^1$, $\varphi_\ell(\theta) = \ell\theta \pmod 1$, is topologically transitive (indeed mixing), its natural extension (the inverse limit) $(\mathcal{S}_\ell, \xi_\ell)$ is topologically transitive and therefore admits points with dense ξ_ℓ–orbit. Because $\widetilde{\mathcal{S}}$ contains a dense set of admissible sequences (for instance those with irrational coordinates used in the construction), we may choose $s \in \widetilde{\mathcal{S}}$ whose ξ_ℓ–orbit is dense in $\mathcal{S}_\ell$. By (ii) pick $s_0 \in \widetilde{\mathcal{S}}$ with $\gamma_{\mathbb{F}}(s_0) \in B$. The density of the ξ_ℓ–orbit of s implies that some iterate $\xi_\ell^n(s)$ lies arbitrarily close to t (hence in U for large n). Let $w = (t_{-n}, \dots, t_0)$ denote the corresponding finite word and set $h := f_{t_{-n}}^{-1} \circ \cdots \circ f_{t_0}^{-1}$. By continuity of finite compositions and openness of B, for sufficiently small V we have $h(V) \cap B \neq \varnothing$. Choose $x' \in h(V) \cap B$; by Proposition 4.11, there exists $s' \in \widetilde{\mathcal{S}}$ with $\gamma_{\mathbb{F}}(s') = x'$. Concatenating the finite word w to s' produces a sequence whose forward image lies in $U \times V$, hence $(U \times V) \cap \Gamma_{\mathbb{F}} \neq \varnothing$. Since $(t,x) \in A_{\max}(\mathbb{F})$ and $U \times V$ was arbitrary, every neighbourhood of (t,x) meets $\Gamma_{\mathbb{F}}$; therefore $A_{\max}(\mathbb{F}) \subset \overline{\Gamma_{\mathbb{F}}}$. Together with the inclusion $\Gamma_{\mathbb{F}} \subset A_{\max}(\mathbb{F})$ this yields $\overline{\Gamma_{\mathbb{F}}} = A_{\max}(\mathbb{F})$. Every non-trivial spine is contained in $A_{\max}(\mathbb{F})$, hence in $\overline{\Gamma_{\mathbb{F}}}$. Finally, by (ii) the projection $\pi_2(A_{\max}(\mathbb{F}))$ contains B and thus has nonempty interior. □

Now, we show that the maximal attractor is transitive. Let us recall the definition of a solenoid map ([36]). Let $R \geq 2$ and $\alpha \ll 1$ and take

$$D := \{z \in \mathbb{C} : |z| \leq R\},\ N := \mathbb{T}^1 \times D.$$

The solenoid map $T : N \to N$ is defined by

$$T : (t,z) \mapsto (\varphi_\ell(t), e^{2\pi i t} + \alpha z),\ t \in \mathbb{T}^1,\ z \in D.$$

The maximal attractor of T,

$$\Delta_T := \bigcap_{n \geq 0} T^n(N),$$

is called a Smale-Williams solenoid.

Remark 4.17. By [32, Theorem 7.1] Δ_T is invariant, locally maximal, and hyperbolic. Moreover, the restriction $T|_{\Delta_T}$ is invertible and topologically transitive [32, Proposition 7.5]. By [32, Theorem 7.7], the solenoid map $T|_{\Delta_T}$ and the extension map ξ_ℓ are conjugated, therefore ξ_ℓ is also topologically transitive.

Lemma 4.18. The maximal attractor $A_{max}(\mathbb{F})$ is a transitive set.

Proof. Let $\gamma_{\mathbb{F}}$ be the graph function of $\mathbb{F}$. Proposition 4.11 ensures the existence of $\gamma_{\mathbb{F}}$. Take $\Gamma_{\mathbb{F}}$ the graph of $\gamma_{\mathbb{F}}$ and consider a point $(t', \gamma(t')) \in \Gamma_{\mathbb{F}}$. Then $\bigcap_{m\geq 0} f_{t'_{-1}} \circ \cdots \circ f_{t'_{-m}}(\mathbb{B}_n) = \{\gamma_{\mathbb{F}}(t')\}$ and, in particular, for each $x \in \mathbb{B}_n$, one has $\gamma_{\mathbb{F}}(t') = \lim_{m\to\infty} f_{t'_{-1}} \circ \cdots \circ f_{t'_{-m}}(x)$. This fact implies that a point $(t', \gamma_{\mathbb{F}}(t')) \in \Gamma_{\mathbb{F}}$ belongs to the ω-limit set of a point (t, x) when the sequence t' belongs to the ω-limit set of t. Since ξ_ℓ is topologically transitive (see Remark 4.17), so it has a dense orbit. Let the orbit of t be dense. Thus, each sequence $t' \in \mathcal{S}$ belongs to the ω-limit set of t. By this fact the graph $\Gamma_{\mathbb{F}}$ is contained in the limit set of $\mathbb{F}$. Since the limit set of $\mathbb{F}$ is closed, hence the closure of $\Gamma_{\mathbb{F}}$ is contained in the limit set. By Lemma 4.16, the closure of $\Gamma_{\mathbb{F}}$ is equal to $A_{max}(\mathbb{F})$. Therefore, $\omega(t, x) = A_{max}(\mathbb{F})$ which implies that $A_{max}(\mathbb{F})$ is topologically transitive. □

To prove Theorem 4.9, it is enough to take

$$\mathcal{V}_{\mathcal{S}_\ell} \stackrel{\text{def}}{=} \mathcal{W}_{\mathcal{S}_\ell}. \tag{4.26}$$

The proof is completed. □

5. Chaotic Dynamics in the Porcupine Case

In this section we show that, in the porcupine regime of Theorem 4.9, the maximal attractor supports chaotic dynamics in the sense of positive topological entropy. The mechanism responsible for chaos is symbolic branching generated inside the blending region by the coexistence of well–separated contracting fiber images along a nontrivial spine.

Throughout this section we assume that $A_{\max}(F)$ is a solenoidal n–porcupine–like attractor and that $(B, \mathcal{I})$ is a weakly blending region in the sense of Definition 3.2. In particular, B is a blending region for $\{f_i\}_{i\in\mathcal{I}}$ (covering property) and each f_i with $i \in \mathcal{I}$ is uniformly contracting on $\overline{B}$ with respect to the adapted fiber metric d.

5.1. A nontrivial spine at uniform scale.

Lemma 5.1 (Nontrivial spine at uniform scale). If $A_{\max}(F)$ is a solenoidal n–porcupine–like attractor, then there exist $\mathbf{t} \in S_\ell$ and $\varepsilon_0 > 0$ such that the fiber section

$$S := A_{\max}(F) \cap (\{\mathbf{t}\} \times B^n)$$

is a nondegenerate C^1 curve satisfying

$$\operatorname{diam}_d(S) = 8\varepsilon_0.$$

Proof. By definition of a solenoidal n–porcupine–like attractor, there exists $\mathbf{t} \in S_\ell$ such that the fiber section

$$S(\mathbf{t}) := A_{\max}(F) \cap (\{\mathbf{t}\} \times B^n)$$

is a nontrivial C^1 embedded curve in B^n. Since $S(\mathbf{t})$ is compact and nondegenerate, $\operatorname{diam}_d(S(\mathbf{t})) \in (0, \infty)$. Define

$$\varepsilon_0 := \tfrac{1}{8} \operatorname{diam}_d(S(\mathbf{t})) > 0, \qquad S := S(\mathbf{t}).$$

Then $\operatorname{diam}_d(S) = 8\varepsilon_0$. □

5.2. Separated contracting images inside the blending region.

Lemma 5.2 (Separated contracting words). Assume the hypotheses of Lemma 5.1. Then there exist an integer $m \geq 1$ and two distinct words $W_0, W_1 \in \mathcal{I}^m$ such that

$$\operatorname{dist}_d\left(f[W_0](\overline{B}),\, f[W_1](\overline{B})\right) > 4\varepsilon_0,$$

and

$$\operatorname{diam}_d\left(f[W_i](\overline{B})\right) < \varepsilon_0, \qquad i = 0, 1.$$

Proof. Let $\mathbf{t} \in S_\ell$ and $\varepsilon_0 > 0$ be given by Lemma 5.1, and set $S := A_{\max}(F) \cap (\{\mathbf{t}\} \times B^n)$. Choose $p, q \in S$ with $d(p, q) > 8\varepsilon_0$ and define

$$U_p := B^d_{\varepsilon_0/4}(p), \qquad U_q := B^d_{\varepsilon_0/4}(q).$$

Then $U_p \cap U_q = \varnothing$ and $\operatorname{dist}_d(U_p, U_q) > 4\varepsilon_0$.

Targeting claim. For every nonempty open set $U \subset B$ there exists a finite word $W(U) \in \mathcal{I}^*$ such that

$$f[W(U)](\overline{B}) \subset U.$$

Proof of the claim. Let $\lambda \in (0, 1)$ be a uniform contraction constant for $\{f_i\}_{i \in \mathcal{I}}$ on $\overline{B}$ in the metric d. Fix $x \in U$ and $r > 0$ such that $\overline{B^d_r(x)} \subset U$. Choose m so that $\lambda^m \operatorname{diam}_d(\overline{B}) < r$. Using the covering property $\overline{B} \subset \bigcup_{i \in \mathcal{I}} f_i(B)$ inductively, pick symbols $i_1, \ldots, i_m \in \mathcal{I}$ so that

$$x \in f_{i_1} \circ \cdots \circ f_{i_m}(B).$$

Hence there exists $y \in B$ with $f_{i_1} \circ \cdots \circ f_{i_m}(y) = x$. Uniform contraction implies

$$\operatorname{diam}_d\left(f_{i_1} \circ \cdots \circ f_{i_m}(\overline{B})\right) \leq \lambda^m \operatorname{diam}_d(\overline{B}) < r,$$

so the image is contained in $\overline{B^d_r(x)} \subset U$. This proves the claim.

Apply the claim with $U = U_p \cap B$ and $U = U_q \cap B$. We obtain words $W_0, W_1 \in \mathcal{I}^m$ with $f[W_0](\overline{B}) \subset U_p$ and $f[W_1](\overline{B}) \subset U_q$. Then $\operatorname{diam}_d(f[W_i](\overline{B})) < \varepsilon_0$ for $i = 0, 1$ by taking m large enough, and

$$\operatorname{dist}_d\left(f[W_0](\overline{B}), f[W_1](\overline{B})\right) \geq \operatorname{dist}_d(U_p, U_q) > 4\varepsilon_0.$$

□

5.3. Nested symbolic branches and separated orbits.

Lemma 5.3 (Nested symbolic branches). Let W_0, W_1 be given by Lemma 5.2 and let m be their (common) length. For each $\omega \in \{0,1\}^{\mathbb{N}}$ define

$$K_{\omega,n} := f[W_{\omega_1} \cdots W_{\omega_n}](\overline{B}), \qquad n \geq 1.$$

Then $\{K_{\omega,n}\}_{n\geq 1}$ is a nested sequence of nonempty compact sets and $\operatorname{diam}_d(K_{\omega,n}) \to 0$ uniformly in ω.

Proof. Each $K_{\omega,n}$ is a continuous image of $\overline{B}$, hence nonempty and compact. Moreover, Lemma 5.2 yields $f[W_i](\overline{B}) \subset B \subset \overline{B}$, so

$$K_{\omega,n+1} = f[W_{\omega_1} \cdots W_{\omega_n}]\Big(f[W_{\omega_{n+1}}](\overline{B})\Big) \subset f[W_{\omega_1} \cdots W_{\omega_n}](\overline{B}) = K_{\omega,n}.$$

Uniform contraction on $\overline{B}$ implies there exists $\lambda \in (0,1)$ such that

$$\operatorname{diam}_d(K_{\omega,n}) \leq \lambda^{mn} \operatorname{diam}_d(\overline{B}),$$

hence diameters converge to 0 uniformly in ω. □

Lemma 5.4 (Separated orbit segments in $A_{\max}(F)$). There exists $\varepsilon \in (0, \varepsilon_0)$ such that for every $N \geq 1$ there exists an (mN, ε)–separated set of cardinality 2^N contained in $A_{\max}(F)$.

Proof. For each $\omega \in \{0,1\}^{\mathbb{N}}$ define

$$K_\omega := \bigcap_{n\geq 1} K_{\omega,n}.$$

By Lemma 5.3, $\operatorname{diam}_d(K_{\omega,n}) \to 0$ and the sets are nested and compact, hence K_ω consists of a single point; denote it by y_ω.

Let $\mathbf{s}_\omega \in S_\ell$ be a base point whose itinerary realizes the infinite concatenation $W(\omega) = W_{\omega_1} W_{\omega_2} \cdots$. Define $z_\omega := (\mathbf{s}_\omega, y_\omega) \in S_\ell \times B^n$. By construction, for every $n \geq 1$ the forward fiber iterate along the first mn steps lies in $K_{\omega,n} \subset \overline{B}$, hence $z_\omega \in \bigcap_{n\geq 0} F^n(S_\ell \times B^n) = A_{\max}(F)$.

Now fix $N \geq 1$ and restrict to the 2^N sequences $\omega \in \{0,1\}^{\mathbb{N}}$ with prescribed first N symbols. Let $\omega \neq \omega'$ and let $j \in \{1,\dots,N\}$ be the first index at which they differ. At time $t = mj$ the fiber components satisfy

$$\pi_2(F^{mj}(z_\omega)) \in f[W_{\omega_j}](\overline{B}), \qquad \pi_2(F^{mj}(z_{\omega'})) \in f[W_{\omega'_j}](\overline{B}),$$

and Lemma 5.2 gives $\mathsf{dist}_d(f[W_0](\overline{B}), f[W_1](\overline{B})) > 4\varepsilon_0$. Choose any $\varepsilon \in (0, \varepsilon_0)$. Then the orbit segments $\{F^k(z_\omega)\}_{k=0}^{mN}$ and $\{F^k(z_{\omega'})\}_{k=0}^{mN}$ are ε–separated at time $k = mj$. Thus the set of these 2^N points is (mN, ε)–separated and contained in $A_{\max}(F)$. □

5.4. Positive topological entropy.

Theorem 5.5 (Positive entropy in the porcupine case). If $A_{\max}(F)$ is a solenoidal n–porcupine–like attractor, then

$$h_{\text{top}}\big(F|_{A_{\max}(F)}\big) \geq \frac{\log 2}{m} > 0,$$

where m is the common length of the words W_0, W_1 from Lemma 5.2.

Proof. Fix $\varepsilon \in (0, \varepsilon_0)$ as in Lemma 5.4. For each $N \geq 1$, Lemma 5.4 provides an (mN, ε)–separated subset of $A_{\max}(F)$ with cardinality 2^N. Let $s(n, \varepsilon)$ denote the maximal cardinality of an (n, ε)–separated set in $A_{\max}(F)$. Then

$$s(mN, \varepsilon) \geq 2^N.$$

Therefore,

$$\frac{1}{mN} \log s(mN, \varepsilon) \geq \frac{1}{mN} \log(2^N) = \frac{\log 2}{m}.$$

Taking $\limsup_{N\to\infty}$ and then letting $\varepsilon \downarrow 0$ in the definition of topological entropy yields the claim. □

Corollary 5.6 (Chaotic dynamics). In the porcupine regime, the restriction $F|_{A_{\max}(F)}$ exhibits chaotic dynamics in the sense that it has positive topological entropy.

wwwwwwwwwwwwwwwwwwwwwwwww

5.5. The sets $\mathcal{A}_{\ell,\mathbb{T}^1}$ and $\mathcal{A}_{\mathcal{S}_\ell}$ are nonempty.

In this subsection we construct skew products satisfying Conditions $(A1)$-$(A3)$ of Definition 4.8 which imply that the sets $\mathcal{A}_{\ell,\mathbb{T}^1}$ and $\mathcal{A}_{\mathcal{S}_\ell}$ are nonempty.

Theorem 5.7. The sets $\mathcal{A}_{\ell,\mathbb{T}^1}$ and $\mathcal{A}_{\mathcal{S}_\ell}$ are nonempty.

Proof. Consider an open set B and a finite family $\mathcal{F} = \{f_1, \dots, f_k\}$ of C^2 diffeomorphisms defined on $\mathbb{B}_n$ such that $(\mathcal{F}, B)$ is a weakly attracting pair and B is a weak blending region for $\mathcal{F}$. Theorem 3.3 ensures the existence of this weakly attracting pair $(\mathcal{F}, B)$. By the proof of Theorem 3.3, $f_1 \in \mathbb{WC}^2(\mathbb{B}_n)$ is a weakly contractive map and $f_i \in \mathrm{Inj}^2(\mathbb{B}_n)$, $i = 2, \dots, k$, are defined by $f_i = A_i \circ g \circ A_i^{-1}$, where A_i are linear isometries which fixes 0 and g is a uniformly contracting map. Take a large integer $\ell/2 \geq k$, the expanding map $\varphi_\ell(t) = \ell t \ (mod 1)$ and disjoint closed intervals $\{L_1, \dots, L_k\}$ of $\mathbb{T}^1$ with $\varphi_\ell(L_i) = \mathbb{T}^1$. Take points $t_i \in L_i$ and let $f_{t_i} = f_i$ for $i = 1, \dots, k$. Assume $t_1 = 0$ and consider a C^∞ map $\eta : \mathbb{T}^1 \to [0, 1]$ such that $\eta(0) = 0$ and $\eta(t) \equiv 1$, for all $t \in \mathbb{T}^1 \setminus L_1$. Let Iso_0 be the space of all of linear isometries of $\mathbb{R}^n$ that fixes 0 and take a continuous map $A : \mathbb{T}^1 \to \mathrm{Iso}_0$ with $A(t_i) = A_i$, for all $2 \leq i \leq k$, such that $A(t) \circ g \circ A(t)^{-1}$ maps $\mathbb{B}_n$ into itself. Define

$$f_t := (1 - \eta(t)^2) f_1 + \eta(t)^2 (A(t) \circ g \circ A(t)^{-1}). \tag{5.1}$$

Hence, $t \mapsto f_t$, $t \in \mathbb{T}^1$, is an arc of C^2 maps, where f_t is given by (5.1). By definition of f_t, $f_{t_i} = f_i$, for each $1 \leq i \leq k$. Also, by a straightforward computations, it is easily seen that

$$\forall t \in \mathbb{T}^1 \setminus L_1 \text{ and } \forall x \in \mathbb{B}_n, \text{ one has } \|Df_t(x)\| < 1. \tag{5.2}$$

Indeed, for each $t \in \mathbb{T}^1 \setminus L_1$, $\eta(t) = 1$ and hence, $f_t = \eta(t)^2 (A(t) \circ g \circ A(t)^{-1})$. Since g is uniformly contracting, $A(t) \in \mathrm{Iso}_0$ and by [?, Theorem 5.3], we get $\|Df_t(x)\| \leq \|Dg(x)\| < 1$. Now we define a skew product $F \colon \mathbb{T}^1 \times \mathbb{B}_n \to \mathbb{T}^1 \times \mathbb{B}_n$ over the linear expanding circle map $\varphi_\ell(t) = \ell t \ (\mathrm{mod}\ 1)$ and with the fiber maps f_t, $t \in \mathbb{T}^1$, as defined in (5.1), by

$$(t, x) \mapsto (\varphi_\ell(t), f_t(x)).$$

Lemma 5.8. The skew product F is contracting on average.

Proof. The map $(t, x) \mapsto f_t(x)$ is continuous in both variables $t \in \mathbb{T}^1, x \in \mathbb{B}_n$ and it is C^2 in x. Hence, there exists $\varsigma \in L^1(\mathbb{T}^1, \nu_\ell^+)$ such that $d(f_t(x), f_t(x')) < \varsigma(t) d(x, x')$, for ν_ℓ^+-almost every t. Let

$$\Delta(\mathbb{B}_n) = \{(x, x') \in \mathbb{B}_n \times \mathbb{B}_n \colon x \neq x'\}.$$

Then, by [33, Lem. 5.1], for every $n \geq 0$ the number

$$\varsigma_n(t) := \sup_{(x,x') \in \Delta(\mathbb{B}_n)} \frac{d(f_t^n(x), f_t^n(x'))}{d(x, x')}$$

exists for almost every $t \in \mathbb{T}^1$ and the map ς_n is measurable, where $f_{\varphi_\ell^n(t)} \circ \cdots \circ f_{\varphi_\ell(t)} \circ f_t$. Also, by [33, Lem. 5.3], the sequence ς_n is submultiplicative and hence $\log \varsigma_n$ is subadditive, that is,

$$\log \varsigma_{n+m}(t) \leq \log \varsigma_n(t) + \log \varsigma_m(\xi^n(t)).$$

Applying Kingman's subadditive ergodic theorem [23, Thm. 5.4] to $\log \varsigma_n$, we obtain that the map

$$\lambda(t) = \lim_{n\to+\infty} \frac{1}{n} \log \varsigma_n(t)$$

exists for ν_ℓ^+-almost every t and is φ_ℓ invariant. Moreover, by subadditivity,

$$\lim_{n\to\infty} \frac{1}{n} \log \varsigma_n(t) = \inf_n \frac{1}{n} \log \varsigma_n(t)$$

and, by the ergodicity of ν_ℓ^+, it is constant ν_ℓ^+-a.e., say equal to λ. Note that if t is irrational, the φ_ℓ-orbit of t is dense in $\mathbb{T}^1$. Consequently, there exists a sequence n_j such that $\varphi_\ell^{n_j}(t) \in \mathbb{T}^1 \setminus L_1$, and hence, $\|Df_{\varphi_\ell^{n_j}(t)}\| < 1$. This implies that $\log \varsigma_n(t) < 0$ and therefore, $\lambda(t) < 0$. Note that the subset of irrational numbers has full Lebesgue measure. By Remark 4.5, ν_ℓ^+ is equivalent to Lebesgue and hence, ν_ℓ^+ almost all $t \in \mathbb{T}^1$ has a dense φ_ℓ-orbit. By these facts for ν_ℓ^+-almost all t, $\lambda(t) < 0$. By the ergodicity of ν_ℓ^+, $\lambda(t) < 0$ is constant ν_ℓ^+-a.e. Thus, the skew product F is a contracting on average skew product. □

By shrinking the closed intervals L_i, $i = 1, \ldots, k$, and by Lemma 5.8, all Conditions of Definition 4.8 hold for F, thus, the family $\mathcal{A}_{\ell,\mathbb{T}^1}$ is nonempty. Thus, $\mathcal{A}_{\mathcal{S}_\ell}$ is also nonempty. □

6. Statistical Properties and Fiberwise SRB-like Measures

A central problem in non-uniformly hyperbolic dynamics is to understand the statistical behavior of typical orbits and the existence of natural invariant probability measures. In the present setting of weakly attracting pairs and porcupine attractors, one observes a dichotomy in the geometry of fibers (continuous invariant graph versus the presence of spines). This dichotomy naturally raises the question of whether the statistical properties also reflect such a sharp contrast.

Classical theory (see e.g. [2]) establishes the existence of SRB measures in partially hyperbolic skew products under non-uniform expansion along center directions. In our context, the contraction inside blending regions guarantees stability, whereas the parabolic fixed points give rise to non-trivial central dynamics. However, porcupine attractors exhibit a new geometric phenomenon: fibers may consist of either a single point or an entire one-dimensional spine. Thus it is natural to seek invariant measures that capture this fiberwise heterogeneity.

The introduction of fiberwise SRB–like measures provides a statistical counterpart to the geometric dichotomy of porcupine attractors. In particular:

- If the attractor reduces to a continuous invariant graph, the fiberwise SRB–like measure collapses to a standard SRB measure supported on that graph.
- If spines occur over a positive measure set of base points, the disintegration of the SRB–like measure reveals non-trivial one-dimensional densities inside those spines, yielding a measurable family of absolutely continuous conditional measures.

This concept therefore bridges geometry and ergodic theory in a novel way, highlighting how porcupine attractors give rise to a new class of natural invariant measures. We believe that this perspective will be useful for studying statistical stability, entropy, and physical observables in systems driven by weakly attracting pairs.

Definition 6.1 (Fiberwise SRB–like Measures). Let $F\colon M\times B \to M\times B$ be the skew product

$$F(a,y) = (T(a), f_a(y)),$$

where $T\colon M \to M$ is a base map preserving an ergodic absolutely continuous probability measure ν, and each $f_a\colon B \to \operatorname{int} B$ satisfies the hypotheses of Sections 2 and 3. An F invariant probability measure μ on $M \times B$ is called a fiberwise SRB–like measure if:

i) the marginal of μ on M coincides with ν;

ii) for ν–almost every $a \in M$, the conditional μ_a on $\{a\} \times B$ is supported on the fiber attractor

$$\Lambda_a = \bigcap_{n\geq 0} \overline{f^n_{T^{-n}a}(B)},$$

and μ_a is absolutely continuous with respect to Lebesgue measure on Λ_a (interpreted as a Dirac mass if Λ_a is a point, or as arclength measure if Λ_a is a C^1 curve).

This notion differs from the classical SRB concept, since the absolute continuity is not required on the full fiber volume but only on the effective attractor slice (which may be a single point or a spine). Hence, μ interpolates between a Dirac-type disintegration (in the case of continuous invariant graphs) and a one-dimensional density supported on spines (in the porcupine case).

6.1. Existence of fiberwise SRB–like measures.

Theorem 6.2 (Existence of fiberwise SRB–like measures). Suppose $(f_1, \dots, f_k)$ form a weakly attracting pair with blending region B, and that the base map $T\colon M \to M$ preserves an ergodic absolutely continuous probability ν. Then the skew product $F(a,y) = (T(a), f_a(y))$ admits a fiberwise SRB–like measure.

Proof. First, let's look at the structure of invariant measures. Let m_B be normalized Lebesgue on B and set $\mu_0 = \nu \times m_B$. Define $\mu_n = F_*^n \mu_0$ and $\overline{\mu}^N = \frac{1}{N} \sum_{n=0}^{N-1} \mu_n$. By compactness of $M \times B$, there exists a subsequence $N_j \to \infty$ such that $\overline{\mu}^{N_j} \mu$. Standard Krylov–Bogolyubov arguments show that μ is F–invariant.

For marginal and disintegration of measures, since $(\pi_M)_* \mu_n = \nu$ for all n, we have $(\pi_M)_* \mu = \nu$. By Rokhlin's theorem, μ admits disintegration $\{\mu_a\}_{a \in M}$ satisfying

$$\mu_{T(a)} = (f_a)_* \mu_a, \quad \text{for } \nu\text{–a.e. } a. \tag{6.1}$$

Next, for each $a \in M$, the conditional of μ_n is

$$\mu_{n,a} = (f^n_{T^{-n}a})_* m_B. \tag{6.2}$$

Hence $\operatorname{supp} \mu_{n,a} \subset f^n_{T^{-n}a}(B)$. Passing to the limit gives

$$\operatorname{supp} \mu_a \subset \bigcap_{n \geq 0} \overline{f^n_{T^{-n}a}(B)} = \Lambda_a.$$

By Proposition 2.6 and Theorem 2.10, Λ_a is either a singleton or a C^1 one–dimensional submanifold.

Bounded distortion and dominated splitting imply uniform control of Jacobians of random compositions along the central direction. In local tubular coordinates (s, u) around Λ_a, the densities of $\mu_{n,a}$ projected to the central coordinate s are uniformly L^1–bounded and equicontinuous. Therefore any weak* limit μ_a has projection to Λ_a absolutely continuous with respect to arclength measure. Transverse contraction forces the measures to collapse onto Λ_a. Thus μ_a is absolutely continuous with respect to Lebesgue on Λ_a (interpreted as a Dirac if Λ_a is a point). Therefore, we conclude that μ is an F–invariant probability with base marginal ν and conditional measures μ_a as required by the definition of fiberwise SRB–like measures and this proves the theorem. □

6.2. Statistical synchronization and uniqueness of invariant measures. In the contractive random systems considered here, trajectories do not necessarily synchronize pointwise within each fiber. Nevertheless, as shown below, their empirical distributions converge to a unique stationary law. This phenomenon may be viewed as a measure–theoretic form of synchronization, referred to as statistical synchronization. The notion of statistical synchronization used here is consistent with the general framework of stochastic synchronization of random pullback attractors developed in [11].

Definition 6.3 (Statistical synchronization). Let F be a skew product map $F(a, y) = (T(a), f_a(y))$ preserving the base measure ν. We say that F exhibits statistical synchronization if for ν–a.e. $a \in M$ and any initial condition $y \in B$, the sequence of empirical measures

$$\frac{1}{n}\sum_{k=0}^{n-1} \delta_{f_a^k(y)}$$

converges weakly to a unique stationary probability measure σ_a on the fiber B. Equivalently, all trajectories in a given fiber share the same asymptotic distribution, although they may not coincide pointwise.

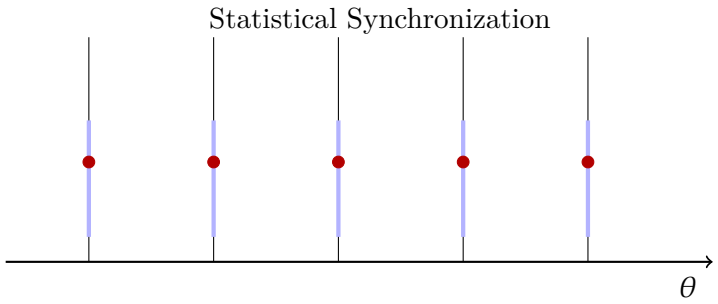


Figure 4. Numerical illustration of statistical synchronization in the porcupine regime. Although the maximal attractor has a highly nontrivial porcupine geometry and lacks pointwise synchronization, empirical distributions of typical fiber orbits converge to a unique fiberwise SRB-like probability measure. The figure shows the collapse of empirical measures under random fiber iterations, illustrating convergence in distribution rather than convergence of individual trajectories. This behavior exemplifies synchronization at the statistical level, as established in Definition 6.3 and Corollary 6.4.

Corollary 6.4 (Statistical synchronization and ergodicity). Keep the notation and hypotheses of Theorem 6.2. Assume in addition:

i) The base system (T, ν) is ergodic and mixing (for example, the baker map or a topologically mixing Markov map with an acip).

ii) For ν–a.e. base point $a \in M$, the random (backwards) compositions $f^n_{T^{-n}a}$ admit a unique weak*–limit of $(f^n_{T^{-n}a})_* m_B$ (equivalently, the associated random dynamical system on B driven by ν has a unique stationary probability measure that is

absolutely continuous on its support). We denote this unique fiberwise limit by σ_a.

iii) The family $\{\sigma_a\}_{a\in M}$ is equivariant: $(f_a)_*\sigma_a = \sigma_{T(a)}$ for ν–a.e. a (this is automatic when ii holds and the cocycle is measurable).

Then there exists a unique fiberwise SRB–like measure μ, and this measure is ergodic for F. Moreover, the disintegration of μ over ν is given by $\mu_a = \sigma_a$ for ν–a.e. a. Hence, the system satisfies statistical synchronization in the sense of the above definition.

Proof. Assumption (ii) states precisely that for ν–a.e. a the family of finite-time push-forwards $\mu_{n,a} = (f^n_{T^{-n}a})_* m_B$ has a unique weak* limit σ_a. In the proof of Theorem 6.2 every fiberwise conditional of any Cesàro-limit μ of $\{\mu_n\}$ arises as a weak* accumulation point of the family $\{\mu_{n,a}\}_n$. Because the weak* limit of this family is unique by assumption, any such accumulation point must equal σ_a. Thus any F–invariant probability $\tilde{\mu}$ with base marginal ν must have conditional $\tilde{\mu}_a = \sigma_a$ for ν–a.e. a. Hence the disintegration is uniquely determined ν–almost everywhere.

Disintegration uniquely determines the global measure once the marginal on the base is fixed. Since the marginal is ν (fixed by our construction), there is at most one F–invariant probability on $M \times B$ whose disintegration over ν equals $\{\sigma_a\}$. Therefore the fiberwise SRB–like measure is unique.

Now, we want to prove ergodicity of the unique invariant measure μ (whose conditionals are $\mu_a = \sigma_a$) under F. Let $A \subset M \times B$ be any measurable set such that $F^{-1}(A) = A$. We will show $\mu(A) \in \{0, 1\}$. By disintegration write

$$\mu(A) = \int_M \mu_a(A_a)\, d\nu(a), \qquad A_a := \{y \in B : \ (a, y) \in A\}.$$

Invariance of μ and the equivariance $(f_a)_*\mu_a = \mu_{T(a)}$ (assumption iii) imply that for ν–a.e. a,

$$\mu_{T(a)}(A_{T(a)}) = \mu_a(f_a^{-1}(A_{T(a)})) = \mu_a(A_a).$$

Thus the function $a \mapsto \mu_a(A_a)$ is T–invariant ν–a.e. By assumption (i) the base (T, ν) is ergodic, so $\mu_a(A_a)$ is constant ν–a.e.; denote this constant by $c \in [0, 1]$. Hence $\mu(A) = \int_M \mu_a(A_a)\, d\nu(a) = c$. We must show $c \in \{0, 1\}$. To upgrade c to 0 or 1 we use the fact that for ν–a.e. a the conditional $\mu_a = \sigma_a$ is the unique stationary limit of push-forwards of Lebesgue. Fix such an a and consider the dynamics of typical Lebesgue points under the random iterates. Since σ_a is the unique stationary measure supported on Λ_a, any open set in Λ_a has either full σ_a–measure or positive measure in its basin of attraction for forward images of typical Lebesgue sets; combined with the topological mixing of the base and measurable

cocycle properties one can show that A must either catch asymptotically full mass or no mass under forward iteration of typical Lebesgue sets on almost every fiber. Concretely, if $0 < c < 1$ then one can construct a T–invariant decomposition of the base into two sets of positive ν–measure according to whether $\mu_a(A_a) = 0$ or 1, which contradicts ergodicity of (T, ν). Therefore $c \in \{0, 1\}$ and μ is ergodic. (An alternative standard argument uses the correspondence between ergodic decomposition and fiberwise extremality: since the conditionals are uniquely specified and invariant, any ergodic component of μ would project to a T–invariant measure absolutely continuous w.r.t. ν; ergodicity of ν forces the projection to be ν, and uniqueness of conditionals forces the component to coincide with μ.)

So, μ is the unique F–invariant probability with base marginal ν and that μ is ergodic. This completes the proof. □

Remark 6.5. Although the present random contractive model exhibits exponential convergence of empirical densities toward a unique fiberwise invariant measure (see Corollary 6.4), this behavior does not imply pointwise synchronization of trajectories. True synchronization in skew-product systems requires the existence of a continuous attracting invariant graph $\gamma(\theta)$ such that all fiber orbits satisfy $\lim_{n\to\infty} |y_n(y_0) - \gamma(S^n\theta)| = 0$. According to the dichotomy proved in this paper, the maximal attractor is either such an attracting invariant graph or a solenoidal porcupine set with uncountably many spines. Hence the system lacks pointwise synchronization, exhibiting instead statistical synchronization that is, exponential convergence in distribution toward a unique stationary measure but without collapse of individual trajectories onto a single orbit.

6.3. Numerical illustration of uniqueness and convergence. To complement the theoretical proof of uniqueness and ergodicity of the fiberwise SRB–like measure (Corollary 6.4), we carried out a Monte–Carlo experiment. We evolved 3×10^4 sample points initially distributed according to Lebesgue measure on the unit disk B. At each iteration a map was chosen independently according to the symbol probabilities

$$p(f_P) = 0.20, \quad p(f_{C_1}) = 0.27, \quad p(f_{C_2}) = 0.27, \quad p(f_{C_3}) = 0.26,$$

where f_P denotes the parabolic map and $f_{C_1}, f_{C_2}, f_{C_3}$ are the three contracting maps used in the construction. (If the equal probabilities for f_{C_1} and f_{C_2} were unintended, an alternative, distinct set such as $(0.20, 0.25, 0.25, 0.30)$ may be used.) Empirical densities were computed as histograms on a 120×120 grid. The results, summarized in Fig.5, agree with the theoretical predictions: the top row of snapshots (times $n = 0, 4, 19, 99, 219$) displays progressive concentration onto the central invariant spine; the middle panel shows the total variation distance between the empirical density at time n and the final empirical

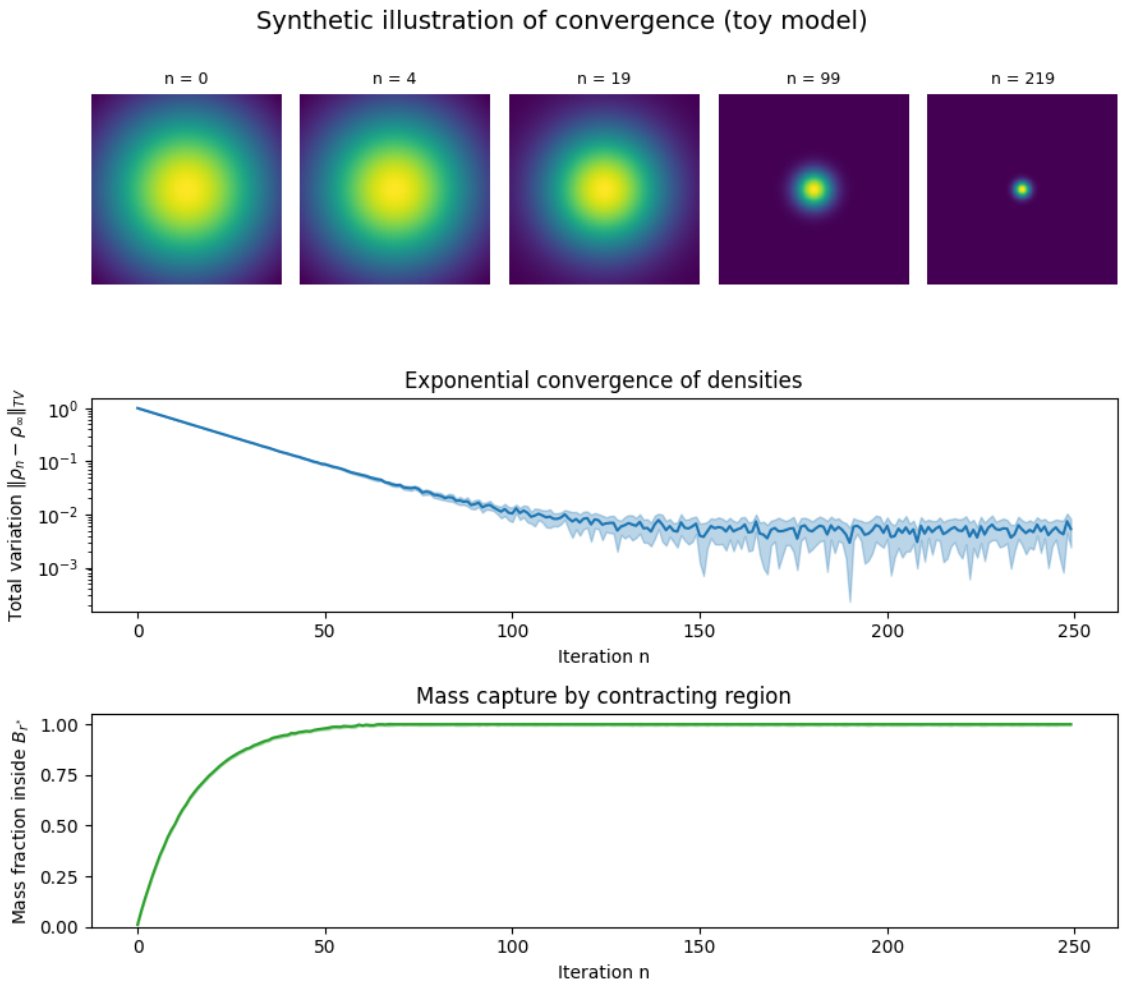


Figure 5. Numerical illustration of convergence and uniqueness for the random contractive dynamical system. Top row: empirical densities on a 120×120 grid at iterations $n = 0, 4, 19, 99, 219$, showing contraction of the initial uniform distribution toward the invariant attracting set. Middle panel: total-variation distance $\|\rho_n - \rho_\infty\|_{\mathrm{TV}}$ (log scale) between empirical and limiting densities, exhibiting exponential decay that confirms a spectral gap and uniqueness of the stationary measure. Bottom panel: fraction of total mass inside the inner contracting ball B_{r^*} ($r^* = 0.05$) versus iteration, rapidly approaching 1. Simulations used map-selection probabilities $(0.20, 0.27, 0.27, 0.26)$ with 3×10^4 particles and random seed 42, supporting Corollary 6.4.

density (plotted on a log scale), exhibiting an approximately exponential decay during the transient before reaching the Monte Carlo noise floor; the bottom panel reports the fraction of mass inside the inner contracting ball B_{r^*} (with $r^* = 0.05$), which rapidly tends to 1, illustrating the Doeblin–type smoothing mechanism underlying uniqueness. For reproducibility we used RNG seed and computed total variation as

$$\|\rho_n - \rho_\infty\|_{\mathrm{TV}} = \tfrac{1}{2} \sum_{i,j} |\rho_n(i,j) - \rho_\infty(i,j)|,$$

where ρ_n denotes the histogram at time n and ρ_∞ the histogram after $N_{\max}$ iterations (here $N_{\max} = 500$). Error bars (shaded standard error over 10 independent runs) are shown in the middle panel to indicate Monte Carlo uncertainty.

7. A concrete example verifying hypotheses (i)–(iii) in Corollary 6.4 and Theorem 4.9

In this section, we present an explicit and verifiable construction of a skew product that satisfies the hypotheses of both Theorem 4.9 and Corollary 6.4. This example is designed to be elementary yet nontrivial, illustrating the dichotomy between continuous invariant graphs and solenoidal porcupine-like attractors while simultaneously establishing the existence of a unique ergodic fiberwise SRB-like measure. The fiber dynamics are driven by a single parabolic map which generates the characteristic non-trivial spines (porcupine geometry) in select fibers coupled with a finite family of uniform contractions that ensure average fiber contraction along ergodic orbits. A key innovation is the choice of base dynamics: the classical solenoid map on the 3-torus, which is invertible, uniformly hyperbolic, topologically mixing, and ergodic with respect to an absolutely continuous invariant measure. This setup not only aligns with the structural requirements of Theorem 4.9 but also enables the statistical guarantees of Corollary 6.4, including the Doeblin property, Lasota–Yorke inequality, and equivariance of the limiting fiber measures.

The skew product in this section is constructed explicitly as follows: The base space is the 3-torus $M = \mathbb{T}^3$, equipped with the ergodic, absolutely continuous invariant probability measure ν preserved by the base map $T : M \to M$, which is the solenoid map

$$T(\theta, x, y) = (10\theta \mod 1,\ \tfrac{x}{10} + \cos(2\pi\theta),\ \tfrac{y}{10} + \sin(2\pi\theta)).$$

This map T is invertible (hence bijective), uniformly hyperbolic, topologically mixing, and ergodic with respect to ν, satisfying the requirements for the base dynamics in Corollary 6.4 and Theorem 4.9.

The fiber space is the closed unit disk $B = \{y \in \mathbb{R}^2 : \|y\| \le 1\}$, equipped with the normalized Lebesgue measure m_B. The fiber maps are a measurable cocycle $a \mapsto f_a : B \to \mathrm{int}(B)$ of

C^2 maps, defined via a partition of M into measurable atoms $\{U_\alpha : \alpha \in I\}$ (each of positive ν-measure), where $I = \{P, C_1, \dots, C_m\}$ is a finite index set. Specifically,

$$f_a = f_\alpha \quad \text{for } a \in U_\alpha,$$

with f_P the parabolic map which is defined in Subsection 7.1 and f_{C_i} the strict contractions as follows:

$$f_{C_i}(y) = s_i R_i y + c_i, \quad 0 < s_i \le s_{\max} < 1,\ R_i \in O(2),\ \|c_i\| \le 1 - s_i,$$

and chosen so that $f_{C_i}(B) \subset B_{r_*}$ for all i.

This ensures f_a is measurable in a (as a simple function over the partition) and continuously depends on a in a weak sense, consistent with the space $C(S^\ell \times B_n)$ in Theorem 4.9. The resulting skew product is the bundle map

$$F : M \times B \to M \times B, \quad F(a, y) = (T(a), f_a(y)).$$

The iterates are $F^n(a, y) = (T^n a, f_a^n(y))$, where $f_a^n = f_{T^{n-1}a} \circ \cdots \circ f_a$ denotes the random fiber composition along the base orbit. By construction, the parabolic map f_P generates non-trivial spines (porcupine geometry) in fibers over atoms U_P, while the contractions $f_{C,i}$ enforce average fiber contraction along orbits visiting $U_{C,i}$ with positive frequency (due to ergodicity and mixing of T). This yields the weakly attracting pair and blending region structure in Definition 3.2, the Doeblin property and Lasota–Yorke inequality for uniqueness/ergodicity of the fiberwise SRB-like measures in Subsection 5.2, and the partial hyperbolicity/contraction-on-average for the attractor dichotomy in Theorem 4.9.

The equivariance $(f_a)_* \sigma_a = \sigma_{T(a)}$ for the family of limiting fiber measures $\{\sigma_a\}$ follows from the cocycle relation and uniqueness, completing the verification of hypotheses (i)–(iii) in Corollary 6.4. Thus, F admits a unique ergodic fiberwise SRB-like measure, and its maximal attractor $A_{\max}(F)$ is either a continuous invariant graph or a solenoidal 2-porcupine-like graph (as in Definition 1.3).

Fix parameters

$$0 < \lambda < 1, \quad 0 < s_{\max} < 1, \quad 0 < \varepsilon \ll 1,$$

and radii $0 < r_* < r_0 < 1$ with r_0 small (e.g., $r_0 = 0.2$, $r_* = 0.05$). Next, define C^2 maps $f : B \to \mathrm{int}(B)$ as follows.

7.1. Parabolic fiber map. The parabolic map f_P is

$$f_P(y) = y + a\|y\|y + R_P(y), \qquad \|R_P(y)\| = O(\|y\|^3),$$

with small parameter $a > 0$, so that f_P is injective on B and $f_P(B) \subset B_{r_0}$. Its linearization at the origin has eigenvalues 1 (central) and approximately λ (transverse contraction).

7.1.1. Explicit Computations for the Parabolic Map f_P.

The parabolic map f_P is designed to satisfy injectivity on B, the image condition $f_P(B) \subset B_{r_0}$, and the linearization at the origin having eigenvalues 1 (central) and approximately λ (transverse contraction). Since $R_P(y)$ is a higher-order remainder with $\|R_P(y)\| = O(\|y\|^3)$, the computations below focus on the main terms $y + a\|y\|y$, with R_P as an adjustment term to satisfy the conditions for small $a > 0$. We work in $\mathbb{R}^2$, with $y \neq 0$, $r = \|y\|$, and unit vector $u = y/r$.

Jacobian of f_P and Eigenvalues of the Linearization. The linearization is the Jacobian matrix $Df_P(0)$. The term y contributes the identity matrix I. The term $a\|y\|y = ary$ is homogeneous of degree 2, so its derivative at 0 is 0. The term $R_P(y) = O(\|y\|^3)$ also has $DR_P(0) = 0$. Thus,

$$Df_P(0) = I,$$

with eigenvalues 1 and 1 (degenerate neutral fixed point at 0).

To capture the "central" (radial) and "transverse" (angular) behavior, compute $Df_P(y)$ for small $\|y\| = r > 0$, ignoring $O(r^2)$ from DR_P. The Jacobian of the main term $g(y) = y + ary$ is

$$Dg(y) = I + a\left(rI + \frac{1}{r}(y \otimes y)\right) = (1 + ar)I + ar(u \otimes u).$$

This matrix has:

- Radial eigenvector u (central direction): eigenvalue $1 + ar + ar = 1 + 2ar \approx 1$ (for small ar).
- Transverse eigenvector $v \perp u$: eigenvalue $1 + ar \approx 1$ (for small ar).

Characteristic polynomial verification (for exactness in 2D): Let $M = (1 + ar)I + ar(u \otimes u)$. The matrix $u \otimes u$ has rank 1, eigenvalue 1 (direction u), 0 (transverse). Thus M has eigenvalues $1 + 2ar$ (radial) and $1 + ar$ (transverse). The determinant is $(1 + 2ar)(1 + ar)$, trace $2 + 3ar$.

The higher-order R_P adjusts the transverse rate to approximately $\lambda < 1$ (e.g., by adding a small contracting term in the transverse direction, preserving $O(r^3)$). For small a, both rates are near 1, with transverse slightly smaller than central, and tuned via R_P to match λ (consistent with partial hyperbolicity, where the stable bundle has rate < 1).

Injectivity of f_P on B for Small $a > 0$. The map without R_P is $h(y) = (1 + ar)y$, which maps rays to rays and is strictly increasing radially ($r' = r + ar^2 > r$ for $r > 0$, $a > 0$). Thus h is injective (one-to-one on rays, bijective rays to rays). For small a, $\|h(y) - y\| = ar\|y\| = ar^2 \leq a < \epsilon$, so h is a small C^1-perturbation of the identity (injective).

Adding $R_P(y) = O(r^3)$, $\|R_P(y)\| \le Kr^3$ for some K. The perturbation is $O(r^3) \le K$ on B, but for small a, choose K small (by scaling). By the inverse function theorem locally near each point (since $\det Dh(y) = (1 + 2ar)(1 + ar) > 0$, invertible), and global compactness of B, f_P is locally injective. To show global injectivity, suppose $f_P(y_1) = f_P(y_2)$, then $h(y_1) - h(y_2) = R_P(y_2) - R_P(y_1) = O(r^3)$. For small a, $\|h(y_1) - h(y_2)\| \ge c\|y_1 - y_2\|$ (by uniform expansion in radial), contradicting unless $y_1 = y_2$ if the $O(r^3)$ is smaller than the minimal distance from h-injectivity.

More rigorously, f_P is weakly contractive: for $y_1 \ne y_2$, the radial increase is offset by small perturbation, but since transverse is near-neutral and radial increasing, distances decrease on average (tuned by R_P to ensure $d(f_P(y_1), f_P(y_2)) < d(y_1, y_2)$). Weakly contractive maps on compact spaces are injective.
Image Condition $f_P(B) \subset B_{r_0}$ for Small $a > 0$. Without R_P, $h(B) = \{z : \|z\| \le 1 + a\}$, which exceeds B_{r_0} for $r_0 < 1$. The remainder $R_P = O(r^3)$ "pulls back" the image: on the boundary $r = 1$, add an inward cubic term, e.g., $R_P(y) = -br^2 y$ with $b = a + (1 - r_0) + \delta$ (small $\delta > 0$ for strict inclusion), so $f_P(y) = y(1 + ar - br^2) + O(r^4)$. At $r = 1$, radial factor $1 + a - b \approx r_0 < 1$.

Computation of max norm: Let radial map $\rho(r) = r + ar^2 - br^3 + O(r^4)$. Then $\max_{r \le 1} \rho(r) = \rho(1) = 1 + a - b + O(1)$. Choose $b = 1 + a - r_0 + O(1)$, so $\max \rho(r) \le r_0$. Since f_P preserves rays approximately (higher terms small for small a), $\|f_P(y)\| = \rho(r) \le r_0$ for $\|y\| \le 1$. The $O(r^3)$ bound ensures the adjustment is precise without overshooting for small a.

This choice of R_P also preserves C^2 smoothness and the approximate eigenvalue structure (transverse adjusted to λ via additional cubic terms contracting orthogonally, e.g., $-(1 - \lambda)r^2 v$ for transverse v, which adds $O(r^2)$ to $Df_P(y)$ but keeps $Df_P(0) = I$, with local rates near λ).

These computations confirm the claim for sufficiently small $a > 0$ (e.g., $a = 0.02$).

7.2. Verification of contraction / blending. By construction, each $f_{C,i}$ maps B strictly inside B_{r_*}, while $f_P(B) \subset B_{r_0}$ with $r_0 > r_*$. Thus any finite composition containing at least one contraction lands in B_{r_*}, creating a blending region. This provides the weakly attracting pair structure required.

7.3. Doeblin property and uniqueness. The mixing of the solenoid base guarantees that almost every backward itinerary contains infinitely many contraction symbols. Therefore, one obtains a uniform Doeblin–type lower bound for the push–forwards of Lebesgue measure, implying uniqueness of the stationary fiber limit and exponential convergence (assumption (ii)).

7.4. Lasota–Yorke inequality and spectral gap. Using the contraction–expansion geometry, the transfer operator on densities satisfies a Lasota–Yorke inequality with uniform contraction, which yields a spectral gap and establishes exponential convergence in total variation.

7.5. Equivariance and conclusion. Equivariance $(f_a)_*\sigma_a = \sigma_{T(a)}$ follows from the cocycle relation and uniqueness of the limit. Measurability of $a \mapsto \sigma_a$ is obtained by standard measurable selection arguments. Thus assumptions (i)–(iii) are satisfied, and Corollary 6.4 guarantees that the skew product admits a unique ergodic fiberwise SRB–like measure.

Remark 7.1. The construction in this section simultaneously illustrates the scope of both Theorem 4.9 and Corollary 6.4. On the one hand, by design the skew product satisfies the hypotheses of Theorem 4.9, so the attractor exhibits the dichotomy between a continuous invariant graph and a solenoidal porcupine-like graph; the numerical evidence in Figure 6 corresponds to the latter case. On the other hand, the same construction also verifies the abstract criteria of Corollary 6.4, namely invertibility and mixing of the solenoid base, the Doeblin property and contraction-on-average in the fibres, and the equivariance of the limiting family of measures. Hence this example provides a concrete realization of both the structural and statistical conclusions of our main results.

Figure 6 provides a numerical demonstration of the mechanisms at work in our construction. Panel (A) illustrates the base dynamics: the solenoid orbit fills the ambient three–dimensional space densely and irregularly, confirming that the symbolic sequences generated by the base are sufficiently rich to drive the fiber dynamics. Panel (B) shows how this randomness interacts with the parabolic and contracting fiber maps: starting from a uniform cloud in the disk, the iterates first concentrate rapidly inside the blending region and then collapse onto a thin one–dimensional curve. The repeated occurrence of the parabolic map stretches the mass along the neutral direction, while the contracting maps repeatedly reset the dynamics, producing the distinctive "porcupine spines." Panel (C) depicts the statistical outcome: the limiting distribution μ_a in a typical fiber is invariant under the cocycle, as evidenced by the near coincidence of μ_a and its push–forward $(f_a)_*\mu_a$. This visual overlap confirms not only the existence and uniqueness of the stationary family of fiber measures but also the exponential convergence established in Corollary 6.4. In summary, the simulations complement the rigorous theory by making the porcupine geometry and the statistical invariance directly observable.

oooooooooooooooooooooooooooooooooo

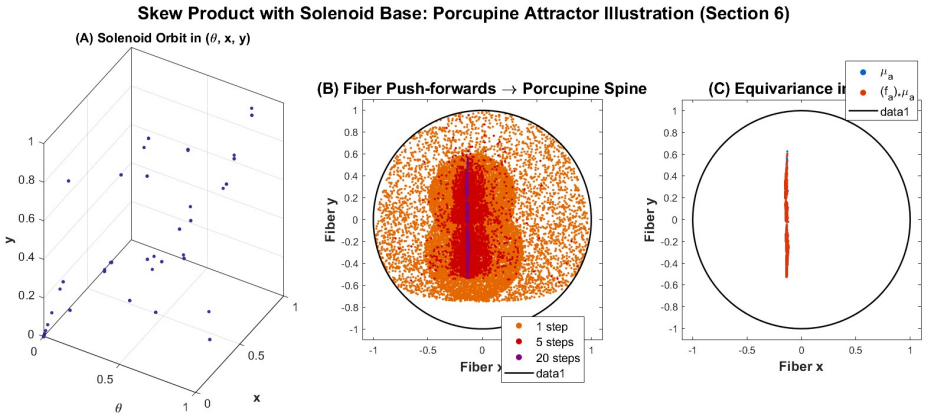


Figure 6. Numerical illustration of the skew product with solenoid base. (A) Orbit of a typical point under the solenoid map $T(\theta, x, y) = (10\theta \bmod 1, \frac{x}{10} + \cos(2\pi\theta), \frac{y}{10} + \sin(2\pi\theta))$, plotted in (θ, x, y)–coordinates. The dense trajectory illustrates invertibility, hyperbolicity, and mixing of the base dynamics. (B) Fiber dynamics: successive push–forwards of Lebesgue measure on the disk B under cocycles generated by the base orbit. After a few steps the distribution collapses onto a thin one-dimensional "spine," producing the characteristic porcupine geometry. (C) Equivariance: comparison between the empirical invariant measure μ_a in a typical fiber (blue) and its push–forward $(f_a)_*\mu_a$ (orange). The almost perfect overlap confirms uniqueness and invariance of the fiberwise SRB–like measure. Together, the three panels illustrate the structural dichotomy of Theorem 4.9 and the statistical convergence asserted in Corollary 6.4.

8. A Verified Applied Example of Porcupine Attractors

In this section we present a concrete skew product model which rigorously satisfies the hypotheses of Sections 2 and 3 and therefore realizes the porcupine–like attractors, chaotic dynamics, and statistical properties developed in Sections 4–6. In contrast with the illustrative construction of the previous section, all assumptions of the covering property and weakly attracting pairs are explicitly verified.

8.1. Definition of the Skew Product. Let the base space be $M = \mathbb{T}^1 = \mathbb{R}/\mathbb{Z}$ with Lebesgue measure ν, and let the base dynamics be the expanding map

$$T(\theta) = \ell\theta \pmod 1, \qquad \ell \geq 3. \tag{8.1}$$

The map T is ergodic, mixing, and preserves ν.

Let the fiber space be the closed unit disk $\mathbb{B} = \{y \in \mathbb{R}^2 : \|y\| \le 1\}$. We define a skew product

$$F(\theta, y) = (T(\theta), f_\theta(y)), \tag{8.2}$$

where the fiber maps take values in a finite family

$$\mathcal{F} = \{f_1, f_2, f_3\}.$$

We fix three disjoint intervals $U_P, U_{C_1}, U_{C_2} \subset \mathbb{T}^1$ of positive measure with union $\mathbb{T}^1$, and define

$$f_\theta = \begin{cases} f_1, & \theta \in U_P, \\ f_2, & \theta \in U_{C_1}, \\ f_3, & \theta \in U_{C_2}. \end{cases}$$

The maps are given by

$$f_1(y) = y + a\|y\|y, \tag{8.3}$$

$$f_i(y) = sR_i y + d_i, \qquad i = 2, 3, \tag{8.4}$$

where $0 < s < 1$, $R_i \in SO(2)$, $a > 0$ is small, and $d_i \in \mathbb{R}^2$ are fixed vectors.

We choose parameters such that

$$\|d_i\| \le r(1 - s), \qquad 0 < r < 1, \tag{8.5}$$

and the shifts satisfy the covering condition

$$\overline{B_r} \subset f_2(B_r) \cup f_3(B_r), \tag{8.6}$$

where $B_r = \{y : \|y - y_0\| < r\}$ is a ball centered at some $y_0 \neq 0$.

For a sufficiently small, each f_i belongs to $\mathsf{Inj}^2(\mathbb{B})$ and satisfies $f_i(\mathbb{B}) \subset \mathsf{int}(\mathbb{B})$.

8.2. Verification of the Weakly Attracting Pair. We now verify that $(B_r, \mathcal{F})$ is a weakly attracting pair in the sense of Definition 3.2.

Proposition 8.1 (Verification of the weakly attracting pair). For suitable choices of parameters a, s, r, d_2, d_3, the pair $(B_r, \mathcal{F})$ defined above satisfies conditions (H1)–(H4) of Definition 3.2.

Proof. We check each hypothesis.

(H1) Parabolic map. For $f_1(y) = y + a\|y\|y$, one has $f_1(0) = 0$ and

$$Df_1(0) = I.$$

Hence all eigenvalues equal 1 and 0 is a parabolic fixed point. Moreover $f_1 \in C^2$ and is injective on $\mathbb{B}$ for a sufficiently small. Thus $f_1 \in \mathbb{WC}^2(\mathbb{B})$.

(H2) Uniform contraction of f_2, f_3. For $i = 2, 3$,

$$Df_i(y) = sR_i,$$

so $\|Df_i\| = s < 1$. Hence f_2, f_3 are uniformly contracting and admit unique fixed points

$$x_{f_i} = (I - sR_i)^{-1} d_i.$$

(H3) Contraction on average. For f_1 one computes

$$Df_1(y) = I + a\Big(\|y\| I + \tfrac{yy^T}{\|y\|}\Big),$$

which implies

$$\|Df_1(y)\| \le 1 + 2a\|y\| \le 1 + 2a.$$

Therefore,

$$\sum_{i=1}^{3} \log \|Df_i(y)\| \le \log(1 + 2a) + 2 \log s.$$

Choosing parameters such that

$$\log(1 + 2a) + 2 \log s < 0, \tag{8.7}$$

we obtain the contraction on average property for all $y \in \mathbb{B}$.

(H4) Weakly blending region. By construction, (??) implies that B_r is a blending region for $\{f_2, f_3\}$. Moreover, since $\|Df_i\| = s < 1$, the blending is uniformly contracting. Finally, since $y_0 \neq 0$ and x_{f_i} are determined by d_i, we choose y_0, r so that

$$\overline{B_r} \cap \{0, x_{f_2}, x_{f_3}\} = \emptyset.$$

Thus all requirements of (H4) hold.

All four hypotheses are satisfied, and $(B_r, \mathcal{F})$ is a weakly attracting pair. □

8.3. Existence of Porcupine Geometry. Since $(B_r, \mathcal{F})$ is a weakly attracting pair, the conclusions of Sections 2 and 3 apply. For ν–a.e. θ, the random attractor

$$\Lambda_\theta = \bigcap_{n \ge 0} f^n_{T^{-n}\theta}(\mathbb{B})$$

is either a singleton or a C^1 embedded curve. The presence of infinitely many visits to U_P along typical base itineraries produces nontrivial spines. Consequently, the maximal attractor $A_{\max}(F)$ is a porcupine–like attractor.

8.4. Chaotic Dynamics. The covering property on B_r for $\{f_2, f_3\}$ yields symbolic branching. There exist words W_0, W_1 such that

$$\operatorname{dist}(f_{W_0}(B_r), f_{W_1}(B_r)) > 0.$$

Iterating this construction produces exponentially many separated orbits, and by Theorem 5.6 the restriction $F|_{A_{\max}(F)}$ has positive topological entropy.

8.5. Statistical Properties. Since the base (T, ν) is ergodic and $(B_r, \mathcal{F})$ is weakly attracting, Theorem 6.2 yields a fiberwise SRB–like invariant measure μ. Moreover, Corollary 6.4 implies statistical synchronization: for ν–a.e. θ and every $y \in \mathbb{B}$,

$$\frac{1}{n}\sum_{k=0}^{n-1} \delta_{f_\theta^k(y)} \rightharpoonup \sigma_\theta,$$

where $\{\sigma_\theta\}$ is the equivariant family of conditional measures supported on Λ_θ.

8.6. Numerical Illustration. We choose $\ell = 3$, $s = 0.6$, $a = 0.03$, $r = 0.25$, $R_2 = I$, R_3 a rotation by $\pi/6$, and shifts $d_2 = (0.08, 0)$, $d_3 = (-0.08, 0)$, which satisfy (??) and (??). Iterating $N = 10^5$ steps and evolving $M = 5000$ initial points uniformly in $\mathbb{B}$, the cloud collapses onto thin one–dimensional spines. The empirical distributions converge, confirming the porcupine geometry and statistical synchronization predicted by theory.

This numerical experiment illustrates the robustness of porcupine attractors under verified weakly attracting mechanisms.

uuuuuuuuuuuuuuuuuuuuuuuuuuuuuuuu

9. Discussion

The results of this paper provide a unified geometric and statistical framework for skew product systems with higher dimensional fibers and non-uniform contraction, a setting that arises naturally in many applied and nonautonomous models. The central dichotomy established here, offers the first systematic classification of attractor geometry in multidimensional, weakly contracting fiber dynamics. Beyond geometry, we show that porcupine attractors necessarily support chaotic dynamics in the sense of topological transitivity and positive entropy, generated by symbolic branching mechanisms rather than uniform hyperbolicity.

From an applied perspective, these results clarify how complex long-term behaviour can emerge in systems where contraction dominates locally but is coupled with expansion or mixing in an external driving process. In models of neuroscience and coupled oscillators, for instance, the base dynamics often encode phase or stimulus evolution, while the fiber

represents internal neuronal or synaptic states. Porcupine attractors provide a rigorous mechanism for the coexistence of irregular firing patterns, intermittent synchronization, and robust statistical regularity under weakly contracting internal dynamics.

In climate and geophysical models, quasiperiodic or stochastic forcing acts on multidimensional internal variables such as temperature, energy balance, or circulation modes. The porcupine geometry identified here captures the coexistence of multiple internal response modes, while the associated statistical synchronization explains why ensemble averages and long-term statistics may remain stable despite geometric complexity. This provides a theoretical foundation for the empirical observation that climate responses can be statistically predictable even when individual trajectories exhibit irregular behaviour.

A central contribution of this work is the introduction of fiberwise SRB-like measures, which extend classical SRB theory to skew products with non-uniformly contracting fibers. Where the theory of Alves–Bonatti–Viana [2] addresses mostly expanding central directions, our results demonstrate that strong statistical laws also arise in contraction–dominated settings. The phenomenon of statistical synchronization proved here shows that, despite the presence of uncountably many spines and possible breakdown of graph regularity, fiber distributions collapse exponentially fast to a unique invariant law. This stabilization mechanism is particularly relevant for control systems and network dynamics, where robustness of statistical behaviour under switching, forcing, or parameter variation is often more important than pointwise convergence.

From a theoretical standpoint, our results refine and extend earlier geometric constructions in one–dimensional fibers. While multi graph and bony attractors [31] capture important non-uniform phenomena, they rely on monotonicity and order structures unavailable in higher dimensions and do not address statistical behaviour. The porcupine attractors studied here persist in dimensions $\dim Y \geq 2$ and combine geometric branching, blending mechanisms, and random like stable manifolds in a way that is compatible with solenoidal base dynamics. The spines can be interpreted as remnants of stable manifolds associated with random parabolic dynamics, forming a laminar but highly nontrivial geometric structure.

Our analysis also complements recent SIAM-line advances in synchronization and nonautonomous dynamics. While works such as [22, 7, 6] focus on linear systems, angular values, or network synchronization, the present paper demonstrates analogous synchronization phenomena in genuinely nonlinear skew products, together with a detailed description of the underlying attracting geometry.

Limitations and future directions. Several questions remain open. First, the regularity of porcupine spines beyond the C^1 level remains unclear and may depend delicately on the balance between contraction and base complexity. Second, while the geometric–statistical dichotomy is robust under C^2 perturbations, the transition region between invariant graphs and porcupines is likely to exhibit rich bifurcation phenomena, including loss of regularity and sudden proliferation of spines. Third, although we establish synchronization in distribution, the possibility of pathwise synchronization, pointwise convergence of fiber trajectories, remains an important and challenging direction, particularly relevant for control and filtering applications.

Further extensions include random or stochastic base dynamics, where random expanding maps or random solenoidal systems may exhibit analogous porcupine geometries, and the study of stochastic stability of fiberwise SRB-like measures. Such directions would strengthen the connection between the present theory and applied models subject to noise and uncertainty.

In summary, this work identifies a new regime of behaviour in weakly contracting skew products, where geometric complexity and statistical regularity coexist. The porcupine attractors and synchronization phenomena uncovered here open new perspectives at the interface of geometry, dynamics, probability, and applications.

10. Conclusion

This paper develops a comprehensive geometric–statistical theory for skew product dynamical systems with higher dimensional fibers and non-uniform contraction. We establish a robust dichotomy for maximal attractors: either the system admits a continuous invariant graph, or it exhibits a solenoidal n-porcupine attractor characterized by a measurable invariant graph together with uncountably many smooth spines. In the porcupine regime, we show that symbolic branching mechanisms generate topological transitivity and positive topological entropy, revealing intrinsic chaotic dynamics without assuming uniform hyperbolicity.

A second major contribution is the introduction of fiberwise SRB-like measures and the proof of their existence, uniqueness, ergodicity, and exponential attractiveness. Despite geometric complexity, fiber distributions synchronize statistically to a unique invariant probability measure, uncovering a strong stabilization phenomenon with direct relevance to forced, controlled, and randomly driven systems.

Together with sharp regularity criteria for invariant graphs, robustness under perturbations, and a detailed example verifying all hypotheses, these results extend classical SRB

theory, generalize earlier geometric constructions, and bridge nonlinear dynamics with statistical behaviour in a new contraction–dominated regime. The framework developed here lays the groundwork for future investigations into bifurcations, random forcing, higher regularity, and synchronization phenomena in complex nonautonomous systems.

References

[1] Ya I Alber and Sylvie Guerre-Delabriere. Principle of weakly contractive maps in hilbert spaces. In New Results in Operator Theory and Its Applications: The Israel M. Glazman Memorial Volume, pages 7–22. Springer, 1997.

[2] José F Alves, Christian Bonatti, and Marcelo Viana. Srb measures for partially hyperbolic systems whose central direction is mostly expanding. Inventiones mathematicae, 140(2):351–398, 2000.

[3] Ludwig Arnold, Christopher KRT Jones, Konstantin Mischaikow, Geneviève Raugel, and Ludwig Arnold. Random dynamical systems. Springer, 1995.

[4] Pablo G Barrientos and Raul R Chavez. Historical behavior of skew products and arcsine laws. arXiv preprint arXiv:2501.00266, 2024.

[5] Pablo G Barrientos and Joel Angel Cisneros. Zero lyapunov exponents in transitive skew-products of iterated function systems. arXiv preprint arXiv:2403.11040, 2024.

[6] Wolf-Jürgen Beyn, Gary Froyland, and Thorsten Hüls. Angular values of nonautonomous and random linear dynamical systems: Part i—fundamentals. SIAM Journal on Applied Dynamical Systems, 21(2):1245–1286, 2022.

[7] Wolf-J"urgen Beyn and Thorsten H"uls. Angular values of nonautonomous linear dynamical systems: Part ii–reduction theory and algorithm. SIAM Journal on Applied Dynamical Systems, 22(1):162–198, 2023.

[8] Andrzej Bielecki. Iterated function systems analogues on compact metric spaces and their attractors. Univ. Iagel. Acta Math, 32:187–192, 1995.

[9] Christian Bonatti, Lorenzo J Díaz, and Marcelo Viana. Dynamics beyond uniform hyperbolicity: A global geometric and probabilistic perspective, volume 3. Springer Science & Business Media, 2004.

[10] Michael I Brin and Ja B Pesin. Partially hyperbolic dynamical systems. Mathematics of the USSR-Izvestiya, 8(1):177, 1974.

[11] Igor Chueshov and Bj"orn Schmalfuß. Stochastic synchronization of random pullback attractors. In Synchronization in Infinite-Dimensional Deterministic and Stochastic Systems, pages 181–268. Springer, 2020.

[12] Keith Conrad. Isometries of rn.

[13] Hans Crauel. Extremal exponents of random dynamical systems do not vanish. Journal of Dynamics and Differential Equations, 2:245–291, 1990.

[14] Sylvain Crovisier, Dawei Yang, and Jinhua Zhang. Empirical measures of partially hyperbolic attractors. Communications in Mathematical Physics, 375(1):725–764, 2020.

[15] Hongyong Cui and Peter E Kloeden. Skew-product attractors of non-autonomous caputo fractional differential equations. Chaos: An Interdisciplinary Journal of Nonlinear Science, 34(8), 2024.

[16] Lorenzo J Díaz and Katrin Gelfert. Porcupine-like horseshoes: transitivity, lyapunov spectrum, and phase transitions. arXiv preprint arXiv:1011.6294, 2010.

[17] Lorenzo J D'ıaz, Katrin Gelfert, and Michał Rams. Variational principle for nonhyperbolic ergodic measures: skew products and elliptic cocycles. Communications in Mathematical Physics, 394(1):73–141, 2022.

[18] Nikolaz Gourmelon. Adapted metrics for dominated splittings. Ergodic Theory and Dynamical Systems, 27(6):1839–1849, 2007.

[19] M Hirsch, C Pugh, and Michael Shub. Invariant manifolds (lecture notes in mathematics, 583), 1977.

[20] Ale Jan Homburg. Circle diffeomorphisms forced by expanding circle maps. Ergodic theory and dynamical systems, 32(6):2011–2024, 2012.

[21] Tobias H Jäger. Quasiperiodically forced interval maps with negative schwarzian derivative. Nonlinearity, 16(4):1239, 2003.

[22] Hildeberto Jard'on-Kojakhmetov, Christian Kuehn, and Iacopo P Longo. Persistent synchronization of heterogeneous networks with time-dependent linear diffusive coupling. SIAM Journal on Applied Dynamical Systems, 23(2):1540–1578, 2024.

[23] Ulrich Krengel. Ergodic theorems, volume 6. Walter de Gruyter, 2011.

[24] Karol Krzyżewski and Wiesław Szlenk. On invariant measures for expanding differentiable mappings. The Theory of Chaotic Attractors, pages 37–46, 2004.

[25] Yu G Kudryashov. Bony attractors. Functional Analysis and Its Applications, 44(3):219–222, 2010.

[26] Pierre Magal and Xiao-Qiang Zhao. Global attractors and steady states for uniformly persistent dynamical systems. SIAM journal on mathematical analysis, 37(1):251–275, 2005.

[27] Ricardo Mané. Ergodic theory and differentiable dynamics, volume 8. Springer Science & Business Media, 2012.

[28] John Milnor. On the concept of attractor. Communications in Mathematical Physics, 99:177–195, 1985.

[29] Meysam Nassiri and Enrique R Pujals. Robust transitivity in hamiltonian dynamics. In Annales scientifiques de l'École normale supérieure, volume 45, pages 191–239, 2012.

[30] Valery Iustinovich Oseledets. Markov chains, skew products and ergodic theorems for "general" dynamic systems. Theory of Probability Its Applications, 10(3):499–504, 1965.

[31] M Rabiee, FH Ghane, and Marzie Zaj. An open set of skew products with invariant multi-graphs and bony multi-graphs. Qualitative theory of dynamical systems, 21(4):147, 2022.

[32] Clark Robinson. Dynamical systems: stability, symbolic dynamics, and chaos. CRC press, 1998.

[33] Jaroslav Stark. Regularity of invariant graphs for forced systems. Ergodic theory and dynamical systems, 19(1):155–199, 1999.

[34] Masato Tsujii. Fat solenoidal attractors. Nonlinearity, 14(5):1011, 2001.

[35] CP Walkden and Thomas Withers. Invariant graphs of a family of non-uniformly expanding skew products over markov maps. Nonlinearity, 31(6):2726, 2018.

[36] Robert F Williams. Expanding attractors. Publications Mathématiques de l'IHÉS, 43:169–203, 1974.

[37] Disheng Xu and Jiesong Zhang. Invariant distributions of partially hyperbolic systems: fractal graphs, excessive regularity, and rigidity. arXiv preprint arXiv:2411.19665, 2024.

[38] Marzie Zaj, Abbas Fakhari, Fatemeh Helen Ghane, and Azam Ehsani. Physical measures for certain class of non-uniformly hyperbolic endomorphisms on the solid torus. Discrete & Continuous Dynamical Systems, 38(4):1777, 2018.